

AIR POLLUTION

Module 6: Air Pollution (3 Lectures)

Definition, sources of air pollution, types air pollutants, atmospheric stability, mixing heights, plume types and meteorological parameters, effects of air pollution, control measures of air pollution



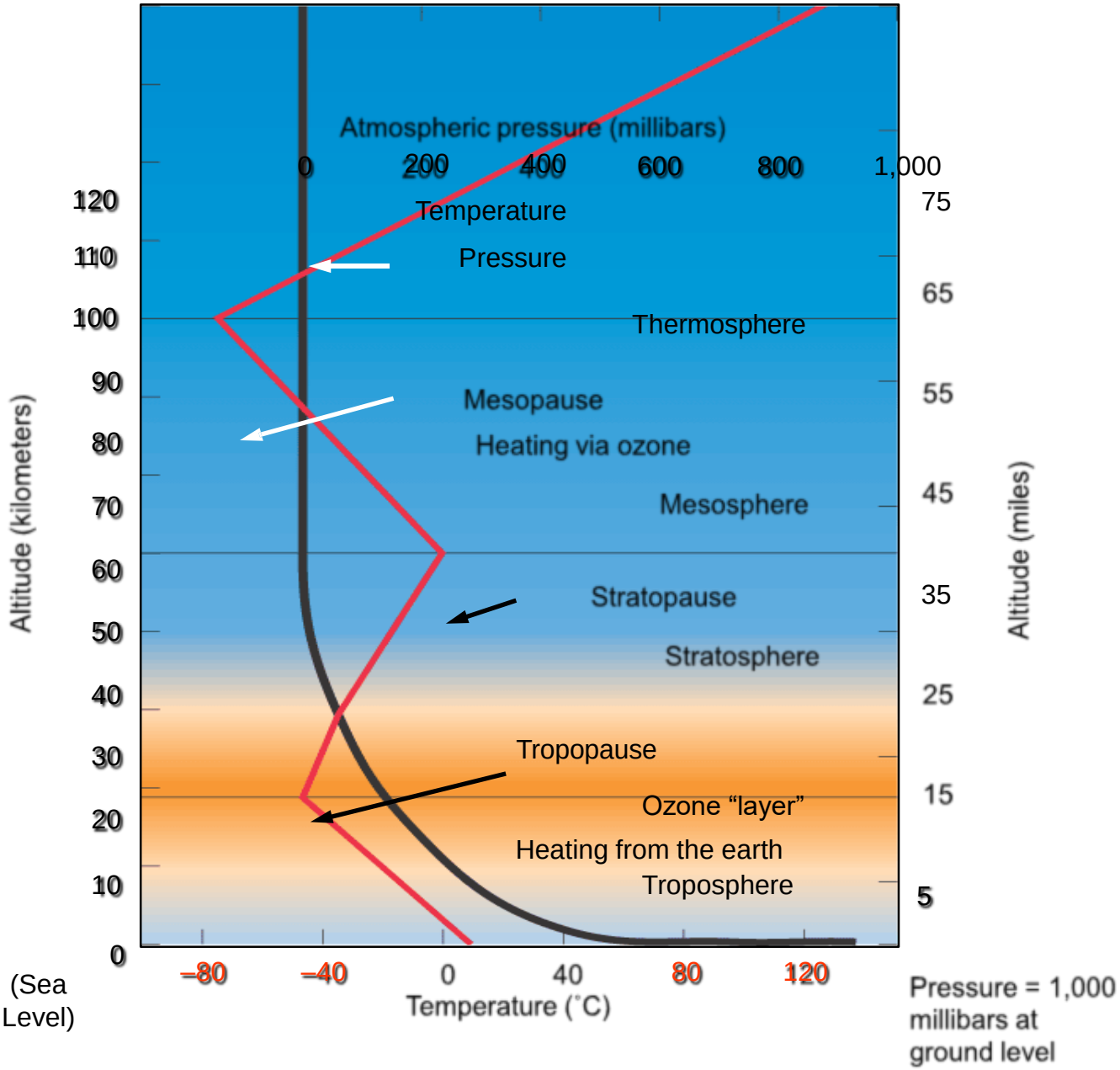
WHAT IS POLLUTION ???

- Pollution is "*change in background conc.*".
- "Any *substance introduced* into the environment that *adversely affects the usefulness* of a resource".
- Pollution happens because no process is 100% efficient; each process produces pollution.

Air pollution

- Air pollution defined as the of *one or more contaminants* or *combinations* in such quantities and of such durations as may be or tend to be *injurious to human, animal or plant life, or property, or which unreasonably interferes with the comfortable enjoyment of life or property or conduct of business*.

The Atmosphere



Types of air pollution

Air pollution can be divided into:

- Indoor air pollution (domestic level)
- Outdoor pollution

Indoor air pollution



Indoor air pollution

- Indoor air pollution is one of the most critical global environmental problems,
- probably exposes more people worldwide to important air pollutants than does pollution in

outdoor air.

- Rural people in developing countries may receive as much as two-thirds (2/3) of global exposure to particulates.

According to WHO…..

- Around 3 billion people still cook and heat their homes using solid fuels in open fires and leaky stoves.
- About 2.7 billion burn biomass (wood, animal dung, crop waste) and a further 0.4 billion use coal.
- Such cooking and heating produces high levels of air pollution with a range of health-damaging pollutants

Indoor air pollution

- In poorly ventilated dwellings, indoor smoke can be 100 times higher than acceptable levels for small particles.

- Exposure is particularly high among women and young children, who spend the most time near the domestic hearth

Pollutant	Sources
Respirable particles	Tobacco smoke Stove Aerosol sprays
Carbon monoxide	Combustion equipment Stove Gas heaters
Nitrogen dioxide	Gas cookers Cigarettes
Carbon dioxide	Combustion Respiration
Formaldehyde	Particle board Carpet adhesives Insulation

Pollutants	Sources
Sulphur dioxide	Coal combustion
Other organic vapours (benzene, toluene, etc)	Solvents Adhesives Resin products Aerosol sprays
Ozone	Electric arcing UV light sources
Radon & “daughters”	Building material
Asbestos	Insulation Fireproofing
Mineral fibres	Appliances

SMOKY COOKING FUELS

Open fire cooking stoves produce heavy smoke containing::

- Fine particles
- Carbon monoxide (CO)
- Polycyclic aromatic hydrocarbons (PAHs)



CARBON MONOXIDE (CO): SOURCES

- Gas, kerosene, wood stoves and coal
- Fires, fireplaces, furnaces
- Leaking chimneys and vents
- Room and water heaters
- Vehicle exhaust in closed garage
- Tobacco smoke

Any place where combustion is incomplete



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Children whose mothers smoke:

- 70% more respiratory problems
- Pneumonia and hospitalization in year 1 **is 38%** higher
- Infant mortality is 80% higher
- 20% of all infant deaths could be avoided if all pregnant stop smoking.
- 5 times higher risk of sudden infant death syndrome (SIDS)

PESTICIDES

Spraying pesticides at home / school:

- Higher concentrations near the floor
- Persistence in some surfaces (carpets, soft toys)
- Overuse and misuse

Children's behaviour and inhalation of pesticides

- Crawling
- Playing close to the floor
- Plush toys
- Hand-to-mouth
- Object-to-mouth

MOSQUITO COILS

- **Household use in Africa, Asia, South America**
- Major active ingredient – pyrethrins
- Long-term exposures linked to asthma and wheezing



SOLVENTS AND VOLATILE ORGANIC COMPOUNDS

Allkanes,, aromattiic hydrocarbons,, allcoholls,, alldehydes,, kettones

- **Sources**:
- Solvents, fabric softeners, and cleaning products
- Paints, glues, waxes and polishing materials
- Spray propellants, dry cleaning fluids
- Pens and markers
- Binders and plasticizers
- Cosmetics: hair sprays, perfumes



BIOLOGICAL POLLUTANTS

Biological pollutants are/were living organisms:

- Animal dander, dust mites, moulds, infectious agents.

Sources of biological agents:

- Water-damaged surfaces and materials
- Humidifiers and stagnant water
- Water vapour from cooking and showering
- Air conditioning systems
- Mattresses, upholstered furniture and carpets

Adverse health effect of indoor air pollutants in health

Acute:

- Irritation of the mucous membranes (eyes, nose, throat)
- Cough, wheeze, chest tightness
- Increased airway responsiveness to allergens
- Increased incidence of acute respiratory illness:
- "cold", pneumonia, otitis media Tracheobronchitis Exacerbation of asthma

Adverse health effect of indoor air pollutants in health

- **Chronic:**
- Long-term exposure decreases lung growth
- Impairment of pulmonary function
- Increased susceptibility to chronic obstructive lung diseases, including asthma
- Others..

Outdoor air pollution



Outdoor air pollution

- Worldwide it is estimated that 1.3 million people -- more than half of them in developing countries -- die every year from urban outdoor air pollution.
- Urban outdoor air pollution is a major environmental health problem affecting people in both developed and developing countries

The main sources of air pollution are:

(a)AUTOMOBILES:

- Motor vehicles are a major source of air pollution throughout the urban areas.
- They emit hydrocarbons, carbon monoxide, lead, nitrogen oxides and particulate matter.
- In strong sunlight, certain of these hydrocarbons and oxides of nitrogen may be converted in the atmosphere into "photochemical" pollutants of oxidizing nature.

The main sources of air pollution are:

(b) INDUSTRIES:

- Combustion of fuel to generate heat and power produces smoke, sulphur dioxide, nitrogen oxides and fly ash.
- Petrochemical industries generate hydrogen fluoride, hydrochloric acid and organic halides.

The main sources of air pollution are:

- Many industries discharge carbon monoxide, carbon dioxide, ozone, hydrogen sulphide and sulphur dioxide.
- Industries discharge their wastes from high chimneys at high temperature and high speed.

The main sources of air pollution are:

(c)DOMESTIC SOURCES :

- Domestic combustion of coal, wood or oil is a major source of smoke, dust, sulphur dioxide and nitrogen oxides.

(d) tobacco smoke:

- The most direct and important source of air pollution affecting the health of many people .
- Even those who do not smoke may inhale the smoke produced by others ("passive smoking").

The main sources of air pollution are:

(e) MISCELLANEOUS:

- These comprise burning refuse, incinerators, pesticide spraying, natural sources (e.g., wind

borne dust, fungi, molds, bacteria) and nuclear energy programmes.

All these contribute to air pollution.

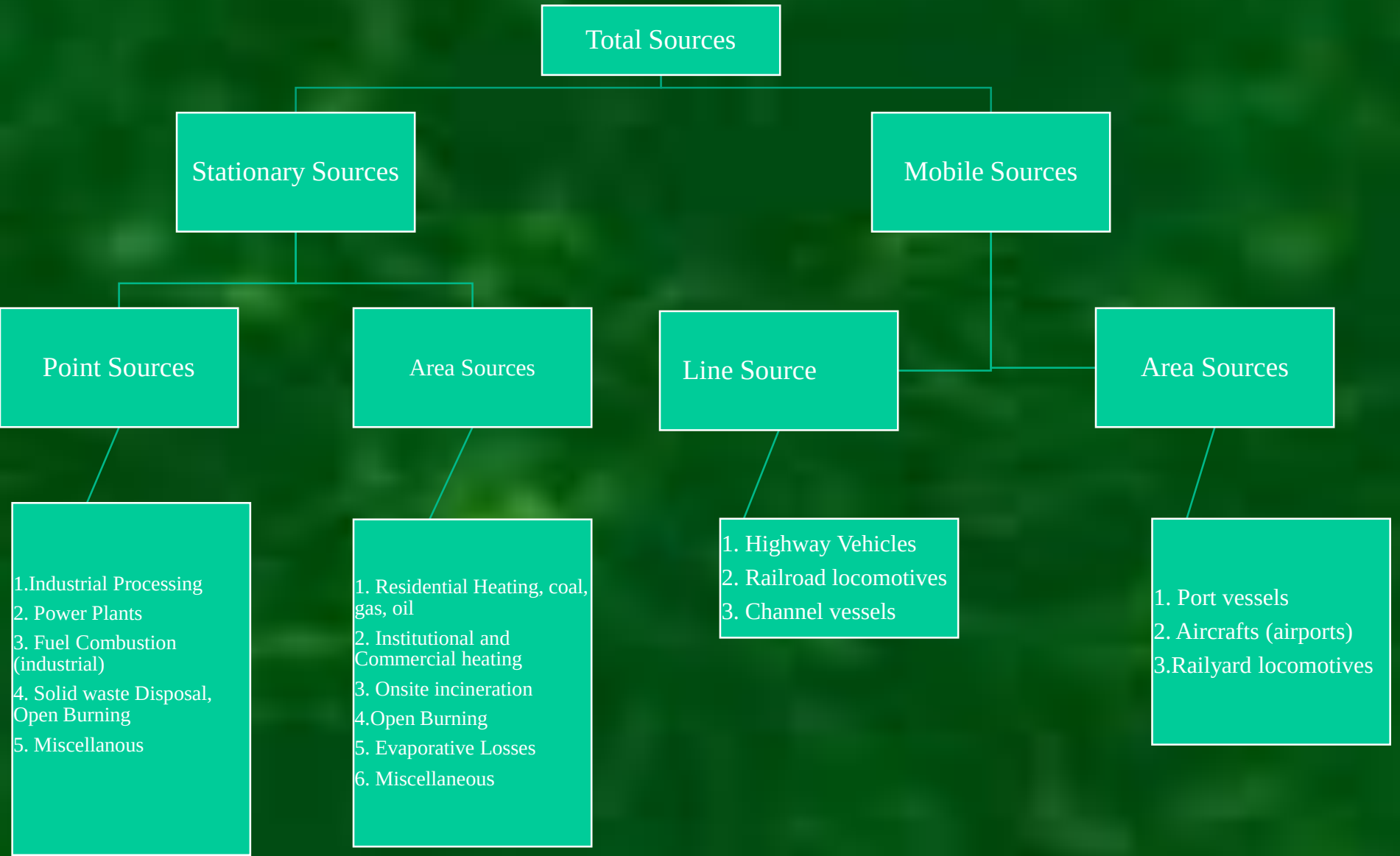
NOTE…

- Although the Earth's atmosphere extends to several layers above the surface, it is only the first30 km that hold the major portion of the atmospheric gases.
- Man is most directly concerned with only the 8-10 km of the atmosphere.
- The level of atmospheric pollution at anyone time depends upon meteorological factors, e.g., topography, air movement and climate.
- Winds help in the dispersal and dilution of pollutants.
- If the topography is dominated by mountains (or tall buildings) the winds become weak and calm, and pollutants tend to concentrate in the breathing zone.

NOTE…

- The vertical diffusion of pollutants depends upon the temperature gradient.
- When there is a rapid cooling of lower layers of air (temperature inversion), there is little vertical motion and the pollutants and water vapours remain trapped at the lower levels and the result is "smog".
- The "temperature inversion "which is more frequent in the winter months than in spring or summer, is a threat to human health.

Types of Air Pollution:



Source and classification



- Natural
- Man- made or anthropogenic

- Natural- pollen grains, volcanic eruptions, forest fires, dust storms, spores, bacteria and other microorganisms.
- Man- made- industrial units, thermal power plants, automobile exhausts, fossil fuel burning, mining, nuclear explosions,

Source and classification

- Stationary
- Mobile

  - Point source- large stationary source
  - Area source- small stationary source and mobile source with indefinite routes
  - Line source- mobile source with definite routes



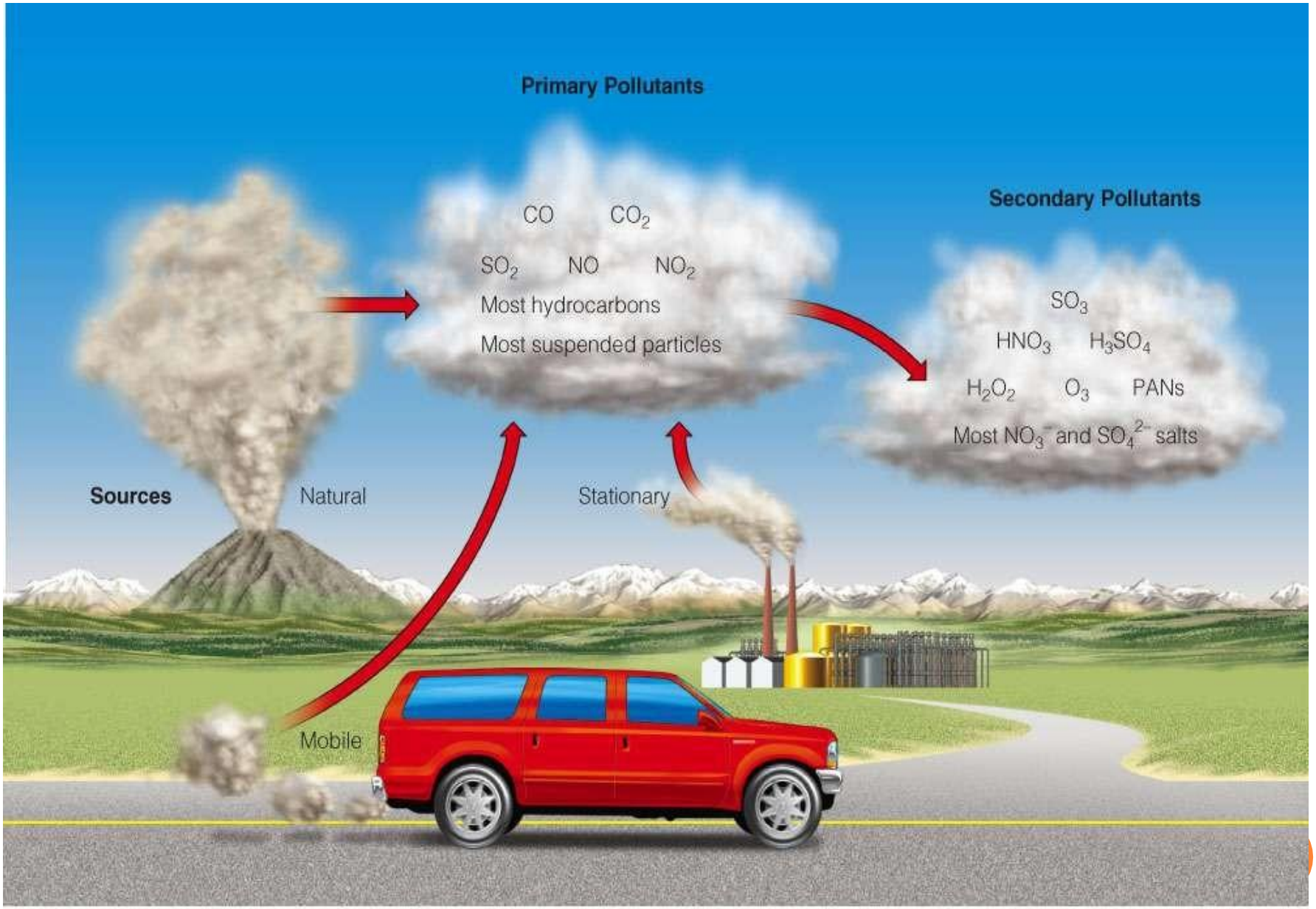
Sources of Air Pollutants:

1) Primary Air Pollutants: those emitted directly from identifiable sources

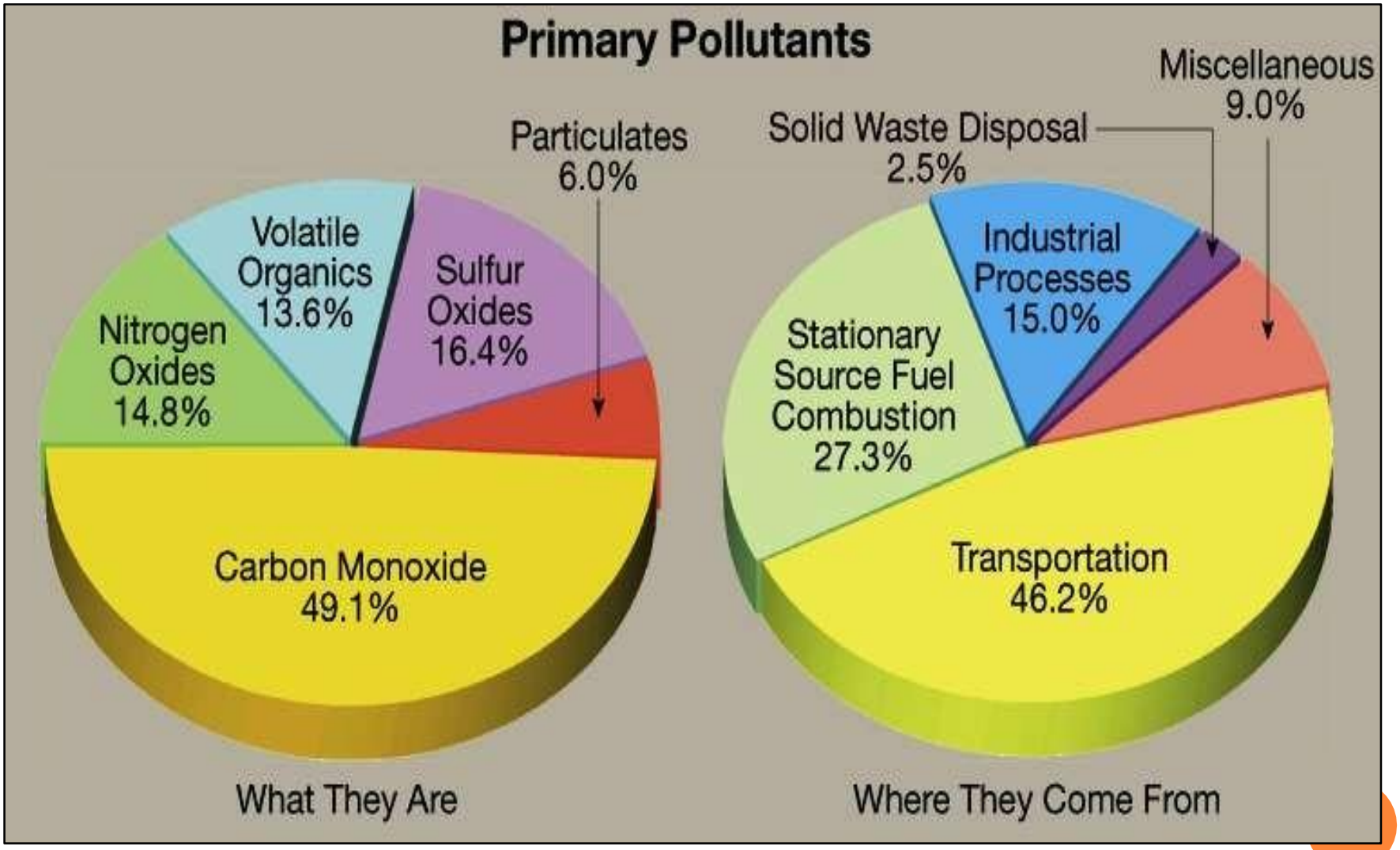
Eg.
- Finer Particles (less than 100 μ in diameter)
- Coarse Particles (greater than 100 μ in diameter)
- Sulphur Compounds
- Oxides of Nitrogen
- Carbon Monoxide
- Halogen Compounds
- Organic Compounds
- Radioactive compounds

2) Secondary Air Pollutants: those which are produced in the air by interaction among two or more primary pollutants

Eg.
- Ozone
- Formaldehyde
- PAN (peroxy acetyl nitrate)
- Photochemical Smog



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Primary Pollutants
Particulates 6.0%
Volatile Organics 13.6%
Sulfur Oxides 16.4%
Nitrogen Oxides 14.8%
Carbon Monoxide 49.1%
What They Are
Miscellaneous 9.0%
Solid Waste Disposal 2.5%
Industrial Processes 15.0%
Stationary Source Fuel Combustion 27.3%
Transportation 46.2%
Where They Come From

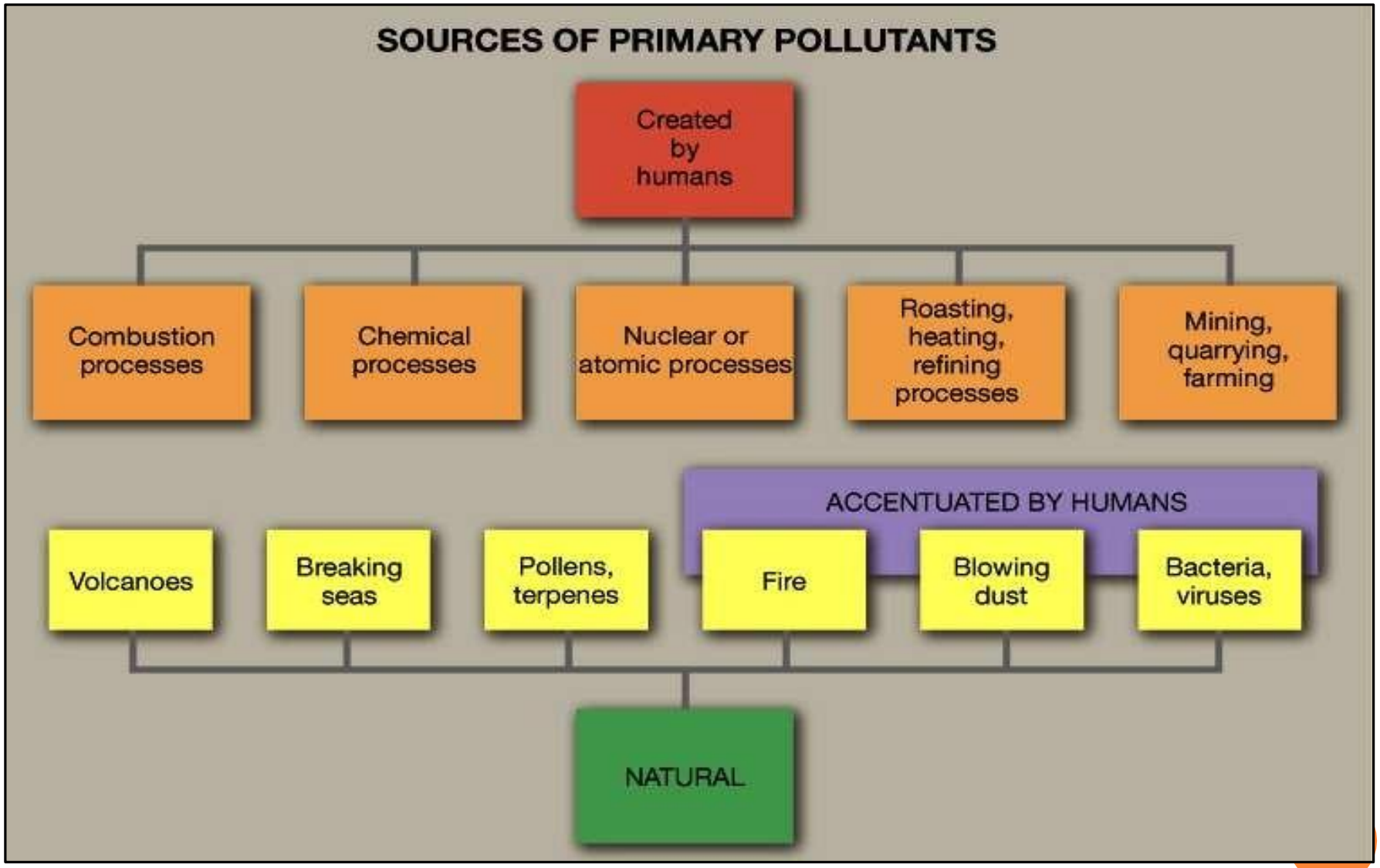
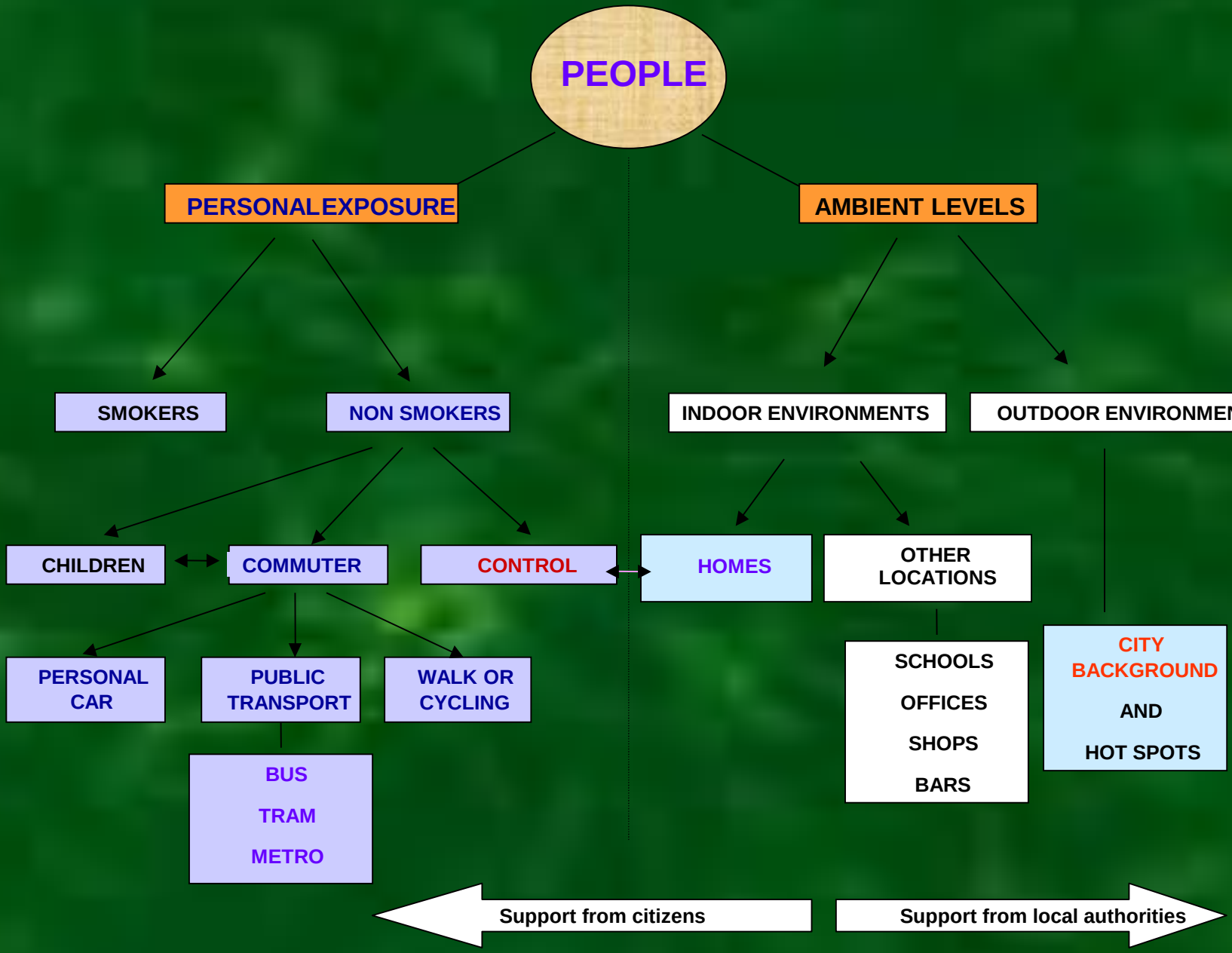

SOURCES OF PRIMARY POLLUTANTS
Created by humans
Combustion processes
Chemical processes
Nuclear or atomic processes
Roasting, heating, refining processes
Mining, quarrying, farming
ACCENTUATED BY HUMANS
Volcanoes
Breaking seas
Pollens, terpenes
Fire
Blowing dust
Bacteria, viruses
NATURAL

Table 1: Sources, Health and Welfare Effects for Criteria Pollutants				
Pollutant	**Description**	**Sources**	**Health Effects**	**Welfare Effects**
Carbon Monoxide (CO)	Colorless, odorless gas	Motor vehicle exhaust, indoor sources include kerosene or wood burning stoves.	Headaches, reduced mental alertness, heart attack, cardiovascular diseases, impaired fetal development, death.	Contribute to the formation of smog.
Sulfur Dioxide (SO_2)	Colorless gas that dissolves in water vapor to form acid, and interact with other gases and particles in the air.	Coal-fired power plants, petroleum refineries, manufacture of sulfuric acid and smelting of ores containing sulfur.	Eye irritation, wheezing, chest tightness, shortness of breath, lung damage.	Contribute to the formation of acid rain, visibility impairment, plant and water damage, aesthetic damage.
Nitrogen Dioxide (NO_2)	Reddish brown, highly reactive gas.	Motor vehicles, electric utilities, and other industrial, commercial, and residential sources that burn fuels.	Susceptibility to respiratory infections, irritation of the lung and respiratory symptoms (e.g., cough, chest pain, difficulty breathing).	Contribute to the formation of smog, acid rain, water quality deterioration, global warming, and visibility impairment.
Ozone (O_3)	Gaseous pollutant when it is formed in the troposphere.	Vehicle exhaust and certain other fumes. Formed from other air pollutants in the presence of sunlight.	Eye and throat irritation, coughing, respiratory tract problems, asthma, lung damage.	Plant and ecosystem damage.
Lead (Pb)	Metallic element	Metal refineries, lead smelters, battery manufacturers, iron and steel producers.	Anemia, high blood pressure, brain and kidney damage, neurological disorders, cancer, lowered IQ.	Affects animals and plants, affects aquatic ecosystems.
Particulate Matter (PM)	Very small particles of soot, dust, or other matter, including tiny droplets of liquids.	Diesel engines, power plants, industries, windblown dust, wood stoves.	Eye irritation, asthma, bronchitis, lung damage, cancer, heavy metal poisoning, cardiovascular effects.	Visibility impairment, atmospheric deposition, aesthetic damage.

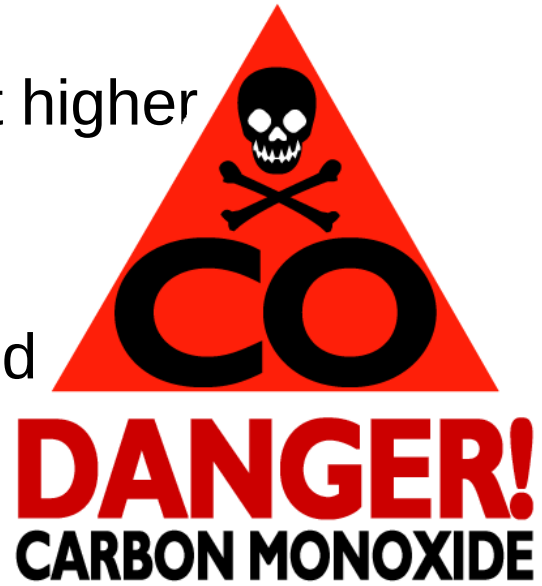
Table-2: Sources, Effects of Air Pollutants on Vegetables

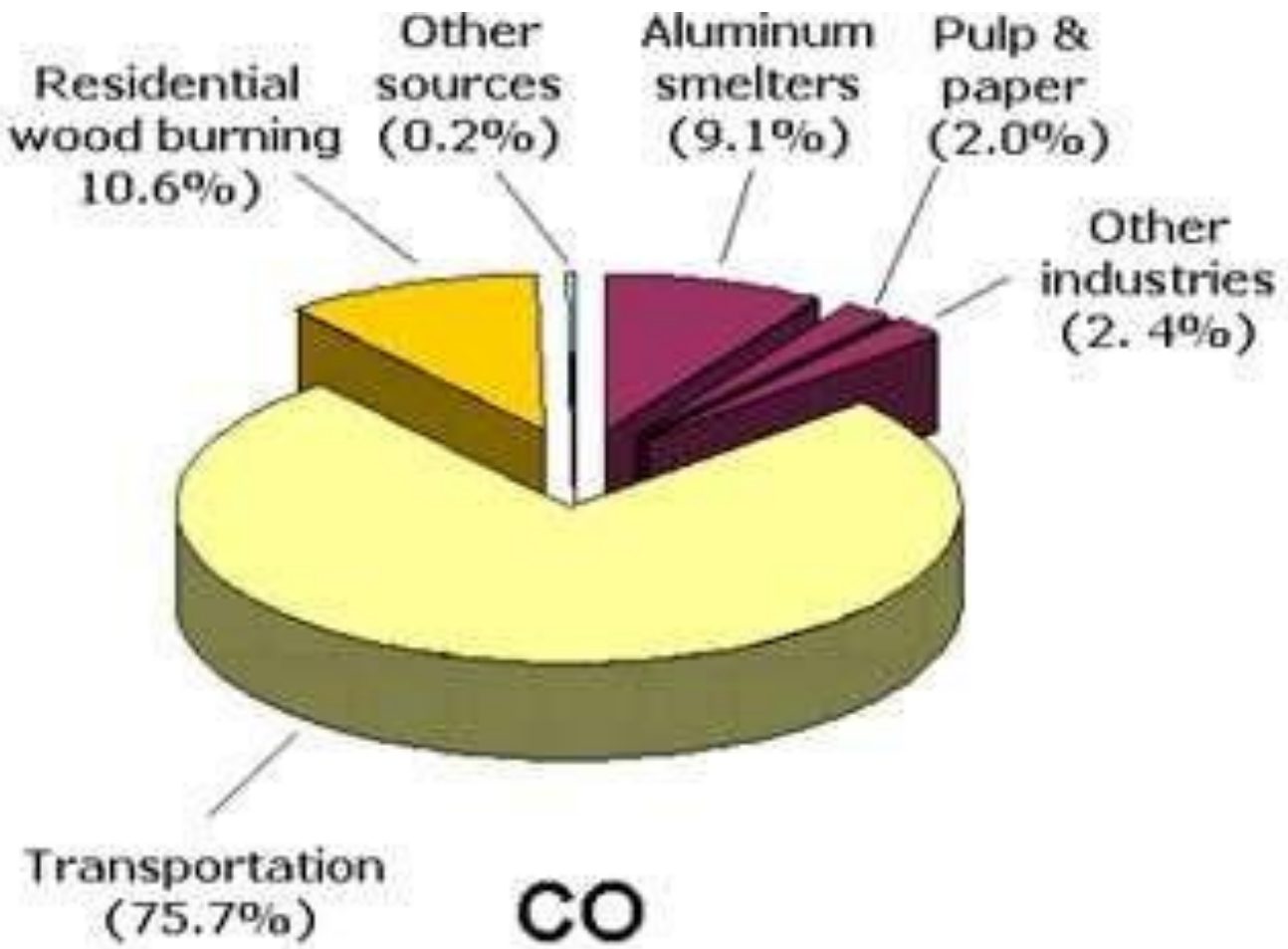
Pollutants	Sources	Effects on Vegetables
Aldehydes	Photochemical reactions	The upper portions of Alfalfa etc. will be affected to Narcosis if 250 ppm of aldehydes is present for 2 hrs duration.
Ozone (O_3)	Photochemical reaction of hydrocarbon and nitrogen oxides from fuel combustion, refuse burning, and evaporation from petroleum products.	All ages of tobacco leaves, beans, grapes, pine, pumpkins and potato are affected. Fleck, stipple, bleaching, bleached spotting, pigmentation, growth suppression, and early abscission are the effects.
Peroxy Acetyl Nitrate (PAN)	The sources of PAN are the same as ozone	Young spongy cells of plants are affected if 0.01 ppm of PAN is present in the ambient air for more than 6 hrs.
Nitrogen dioxide (NO2)	High temperature combustion of coal, oil, gas, and gasoline in power plants and internal combustion engines.	Irregular, white or brown collapsed lesion on intercostals tissue and near leaf margin. Suppressed growth is observed in many plants.
Ammonia & Sulfur dioxide	Thermal power plants, oil and petroleum refineries.	Bleached spots, bleached areas between veins, bleached margins, chlorosis, growth suppression, early abscission, and reduction in yield and tissue collapse occur.
Chlorine (Cl2)	Leaks in chlorine storage tanks, hydrochloric acid mists.	If 0.10 ppm is present for at least 2 hrs, the epidermis and mesophyll of plants will be affected.
Hydrogen fluoride, Silicon tetrafluoride	Phosphate rock processing, aluminum industry, and ceramic works and fiberglass manufacturing.	Epidermis and mesophyll of grapes, large seed fruits, pines and fluorosis in animals occur if 0.001 ppm of HF is present for 5 weeks.
Pesticides & Herbicides	Agricultural operations	Defoliation, dwarfing, curling, twisting, growth reduction and killing of plants may occur.
Particulates	Cement industries, thermal power plants, blasting, crushing and processing industries.	Affects quality of plants, reduces vigor & hardness and interferences with photosynthesis due to plugging leaf stomata and blocking of light.
Mercury (Hg)	Processing of mercury containing ores, burning of coal and oil.	Greenhouse crops, and floral parts of all vegetations are affected; abscission and growth reduction occur in most of the plants.


PEOPLE
PERSONAL EXPOSURE
AMBIENT LEVELS
SMOKERS
NON SMOKERS
INDOOR ENVIRONMENTS
OUTDOOR ENVIRONMENTS
CHILDREN
COMMUTER
CONTROL
HOMES
OTHER LOCATIONS
PERSONAL CAR
PUBLIC TRANSPORT
WALK OR CYCLING
SCHOOLS
OFFICES
SHOPS
BARS
CITY BACKGROUND
AND
HOT SPOTS
BUS
TRAM
METRO
Support from citizens
Support from local authorities

Major Pollutants sources and effects

- Carbon monoxide (CO)- colorless, odorless, tasteless gas.
- No effect at normal conc. (0.1ppm) but higher seriously affect.
- Volcanoes, natural gas emissions, seed contribute to CO.
- Transport sector contribute 75% CO.
- Residential wood burning 10%, industrial process 15% CO.




Residential wood burning 10.6%)
Other sources (0.2%)
Aluminum smelters (9.1%)
Pulp & paper (2.0%)
Other industries (2. 4%)
Transportation (75.7%)
CO

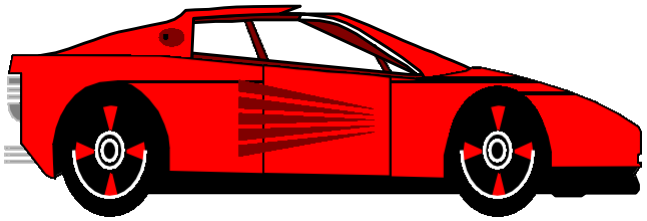
Effects:

- Reduce oxygen carrying capacity of blood.
- Decrease in vision and causes cardio vascular disorders.

- Carbon dioxide ($C0_2$)- Fossil fuel combustion.
- Jet plane use O_2 and release CO_2.
- Burning Effects:
- Causes headache and nausea.
- Effect on climate, increase global temp.



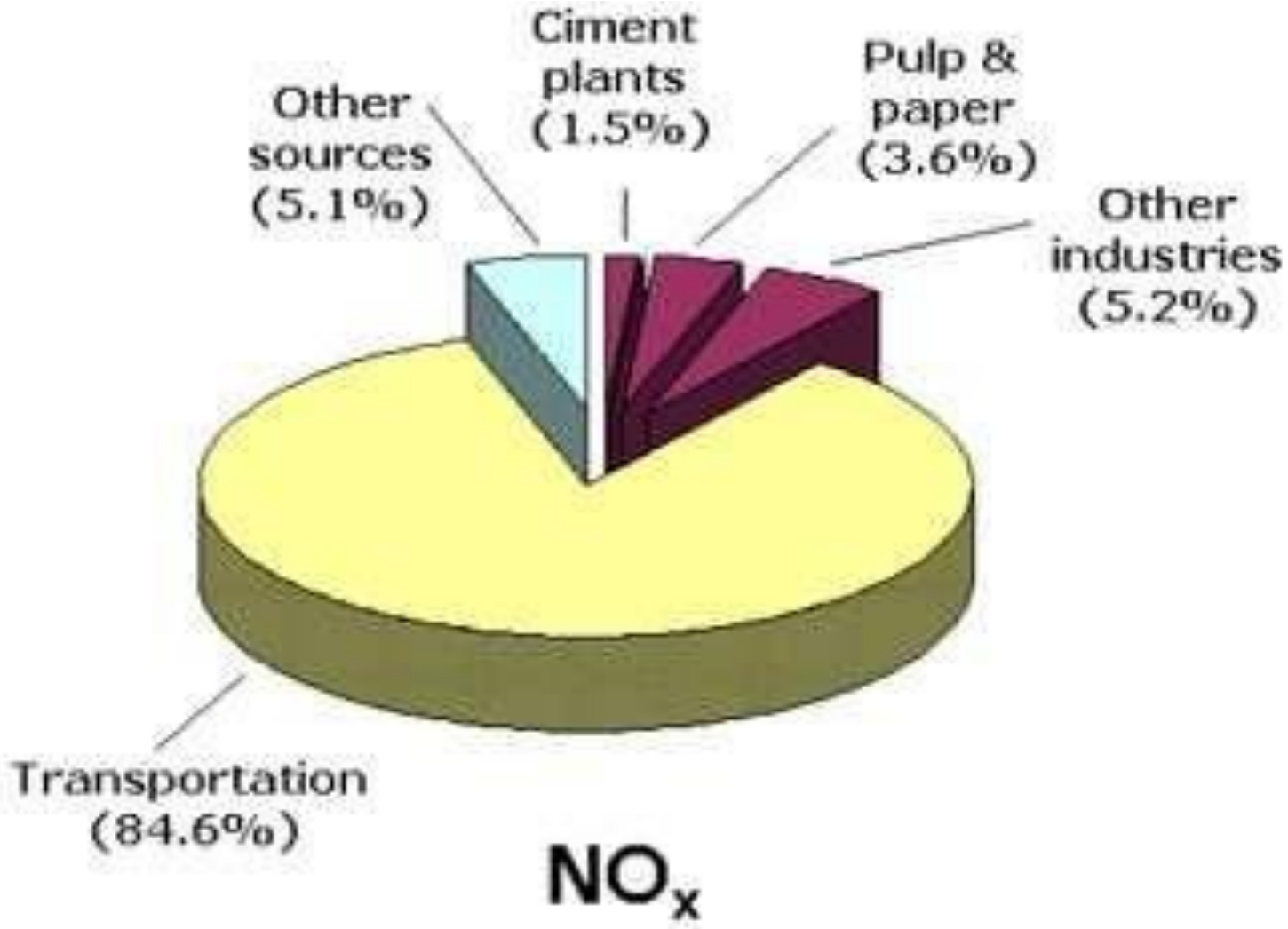
- Oxides of nitrogen – NOx group contains NO, NO_2, N_2O.
- Fuel combustion in automobiles and industries.
- Lightening.
- Forest fires.
- Natural ionizing radiations.



Effects:

- Reduce blood carrying capacity.
- Causes lung problems.



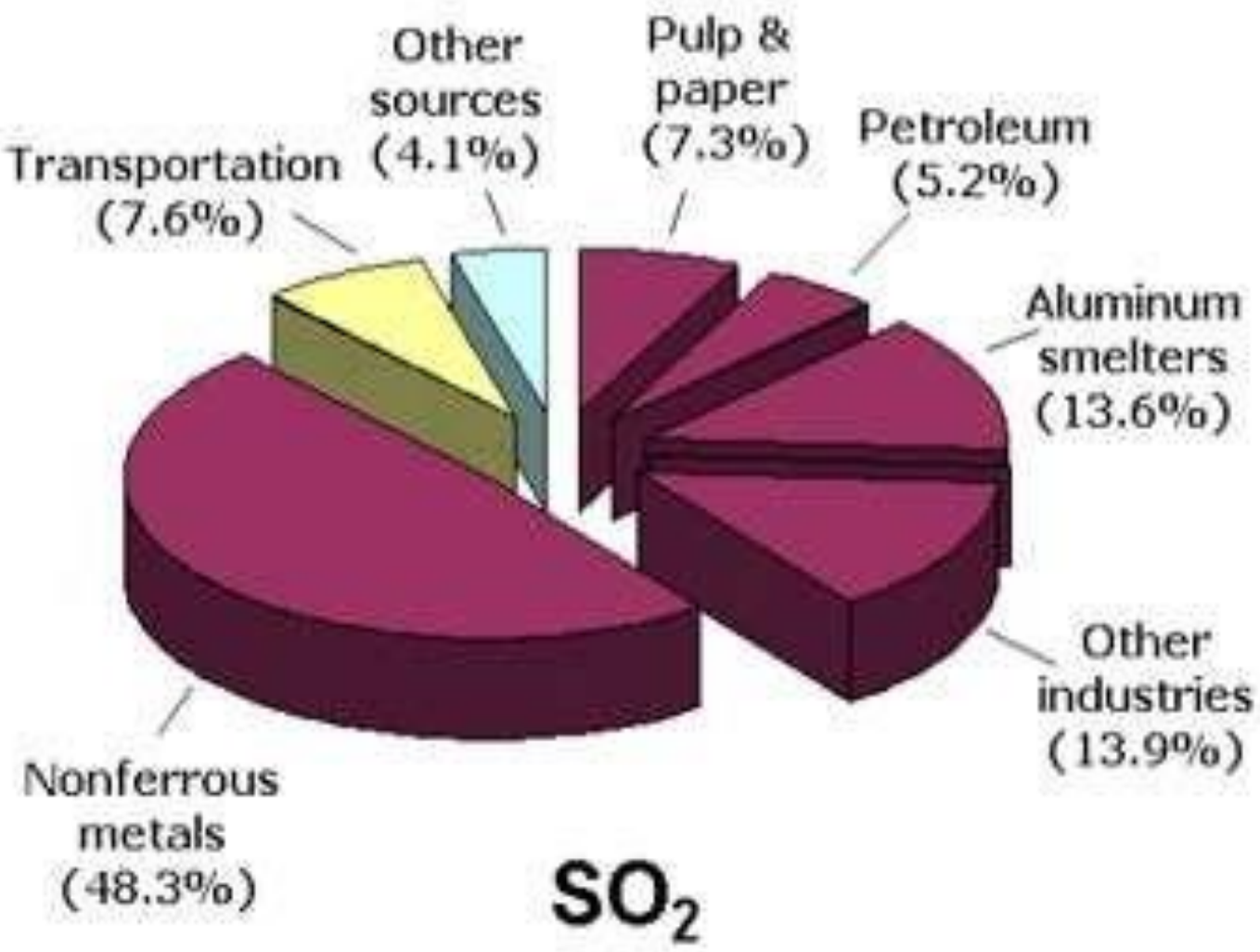

Ciment plants (1.5%)
Pulp & paper (3.6%)
Other sources (5.1%)
Other industries (5.2%)
Transportation (84.6%)
NO_x

- Oxides of sulphur – generally called SOx, include SO_2, SO_3.
- 67% SOx pollution due to volcanic activities and other natural sources.
- Remaining due to fossil fuel burning, transportation.
- Industrial activities.

Effects:

- Respiratory problems
- Marbles, clothes, paper, leather also affected.
- Plants also heavily affected.




Other sources (4.1%)
Pulp & paper (7.3%)
Transportation (7.6%)
Petroleum (5.2%)
Aluminum smelters (13.6%)
Other industries (13.9%)
Nonferrous metals (48.3%)
SO_2

- Hydrocarbons (HC) – these include methane, ethylene, acetylene, terpenes etc.
- Sources include coal fields, natural fires.
- Incomplete combustion
- Forest fires
- Agricultural burning



Effects:

- Carcinogenic effect
- Form ozone and PAN which are harmful.
- Damage plants, rubber materials, fabric and paints.

Particulate matter :

- represents a complex mixture of organic and inorganic substances.
- Mass and composition tend to divide into two principal groups :
- coarse particles larger than 2.5 µm in aerodynamic diameter, and fine particles smaller than 2.5 µm in aerodynamic diameter.
- The smaller particles contain the secondarily formed aerosols ,combustion particles and recompensed organic and metal vapours.
- The large particles usually contain earth's crustal material and fugitive dust from roads and industries.



Particulate matter :

- Particulate matter of respirable size may be emitted from a number of sources, some of them natural (e.g. dust storms)and many others that are more widespread and more important (e.g., power plants and industrial processes, domestic coal burning, industrial incinerators).

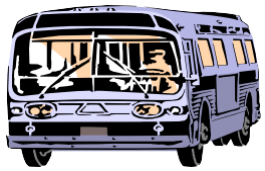
Particulate materials

- Particles of different substances suspended in the air
- In the form of solid particles and liquid droplets
- Particles vary widely in size
- Different particulate materials are aerosols, dust, smoke, fumes, mist, fog, fly ash etc.



- Fine particles come from a variety of sources:
  - -diesel trucks and buses
  - -construction equipment
  - -power plants
  - -woodstoves
  - -wildfires



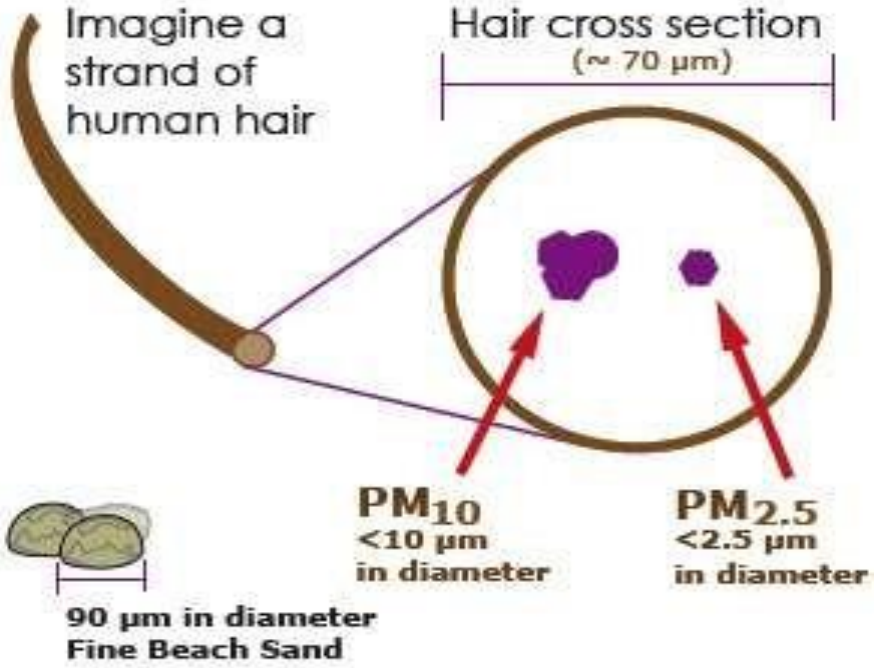


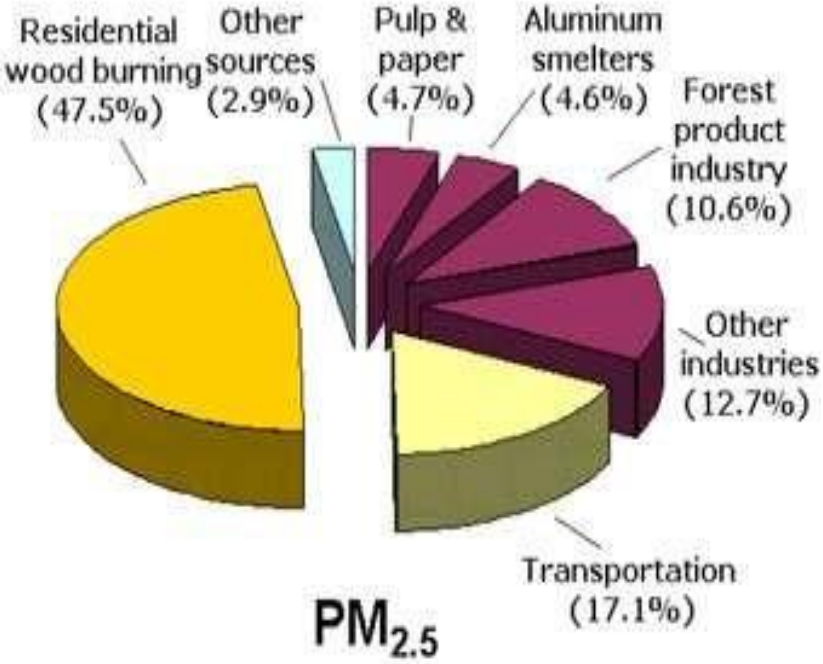




- Also, Chemical reactions in the atmosphere can transform gases into fine particles.

How small is PM?





Effects:

- Premature death
- Aggravated asthma
- Acute respiratory symptoms
- Chronic bronchitis
- Decreased lung function (shortness of breath)
- People with existing heart and lung disease, as well as the elderly and children, are particularly at risk

Ground level ozone

- This is formed when pollutants such as nitrogen oxides and volatile organic compounds (VOCs) react in sunlight .
- High levels can cause breathing problems, reduce lung function and trigger asthma symptoms.
- Ground level ozone can also seriously damage crops and vegetation.
- Ozone is a powerful greenhouse gas and contributes to global warming both directly and by reducing carbon uptake by vegetation.

Ground level ozone

- The WHO guidelines are 150–200 µg/m3 (0.076–0.1 ppm) for one hour exposure and 100–200 µg/m3 (0.05-0.06 ppm) for 8 hour exposures (WHO, 1987a)

SMOG

- Smog is made up of mixture of those pollutants in the atmosphere .
- Combination of words smoke and fog.
- There are two types : reducing smog characterized by sulpher dioxide and particulate.
- And photochemical smog characterized by ozone and other oxidants.

- 1st smog related deaths were in London in 1873; death of 500 people; can you imagine how much worse the atmosphere is now?
- Limits visibility
- Decreases UV radiation
- Yellow/black color over cities
- Causes respiratory problems and bronchial related deaths





- Other pollutant...

Volatile organic compounds:

- VOCs are divided into the separate categories of methane (CH_4) and non-methane (NMVOCs)
- Methane is an extremely efficient greenhouse gas which contributes to enhanced global warming
- NMVOCs, the aromatic compounds benzene, toluene and xylene are suspected carcinogens and may lead to leukaemia through prolonged exposure .

Polynuclear aromatic hydrocarbons (PAH) :

- are a large group of organic compounds with two or more benzene rings.
- They are formed mainly as a result“ of pyrolytic processes, especially the incomplete combustion of organic materials, as well as in natural processes such as carbonization.
- There are about 500 PAH in the air ,the best known is BaP.

Polynuclear aromatic hydrocarbons (PAH) :

- The average total BaP content in the mainstream smoke of one cigarette was 35 ng before 1960and 18 ng in *1978/79;* modern low-tar cigarettes give 10 ng BaP .
- The concentration of BaP in a room extremely polluted with cigarette smoke can be 22 ng/m3.
- In the past, chimney sweeps and tar-workers were dermally exposed to substantial amounts of PAH and there is sufficient evidence that skin cancer in many of these workers was caused by PAH.

Polynuclear aromatic hydrocarbons (PAH) :

- Epidemiological studies in coke-oven workers, coal-gas workers and employees in aluminium production plants provide sufficient evidence of the role of PAH in the induction of lung cancer.
- Owing to its carcinogenicity, no safe level of PAH can be recommended.

Toxic metals:

Lead Cadmium Copper

- Those metals are produced from different factories and can have immediate health effects and long term effects on human health

Others…

- **Chlorofluorocarbons: (CFCs) –**
- **Peroxyacetyl nitrate (PAN)**

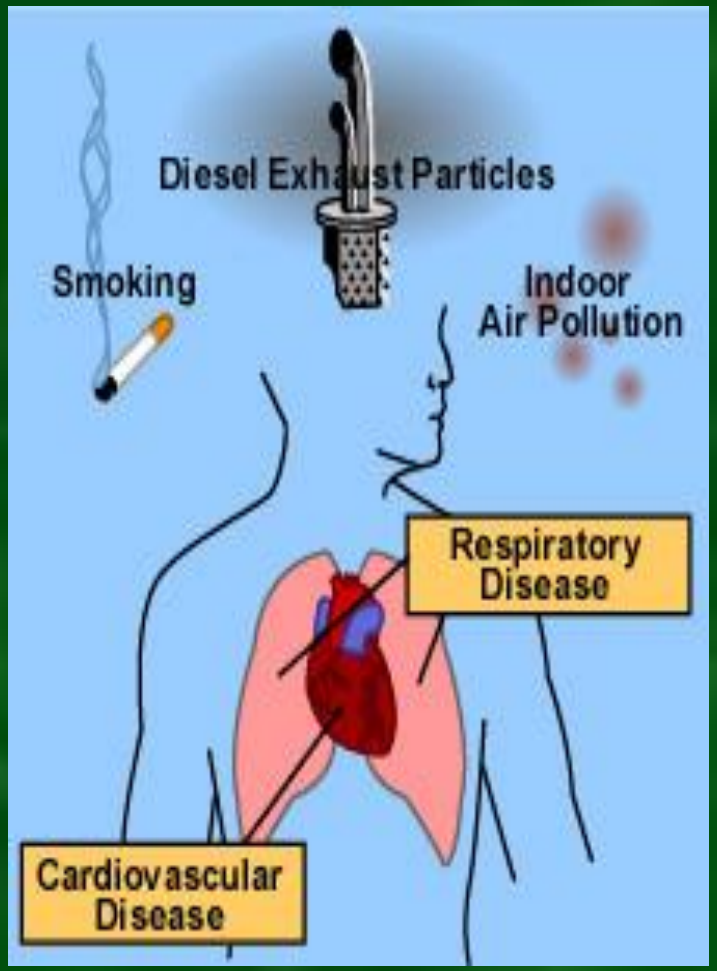
Secondary Pollutants

- Some primary air pollutants react with one another or with other chemicals to form secondary pollutants.

- Atmospheric **sulfuric acid** is one example of a secondary pollutant.
- Air pollution in urban and industrial areas is often called **smog**.
- **Photochemical smog**, a noxious mixture of gases and particles, is produced when strong sunlight triggers **photochemical reactions** in the atmosphere.
- The major component of photochemical smog is **ozone**.

Human Health Effects

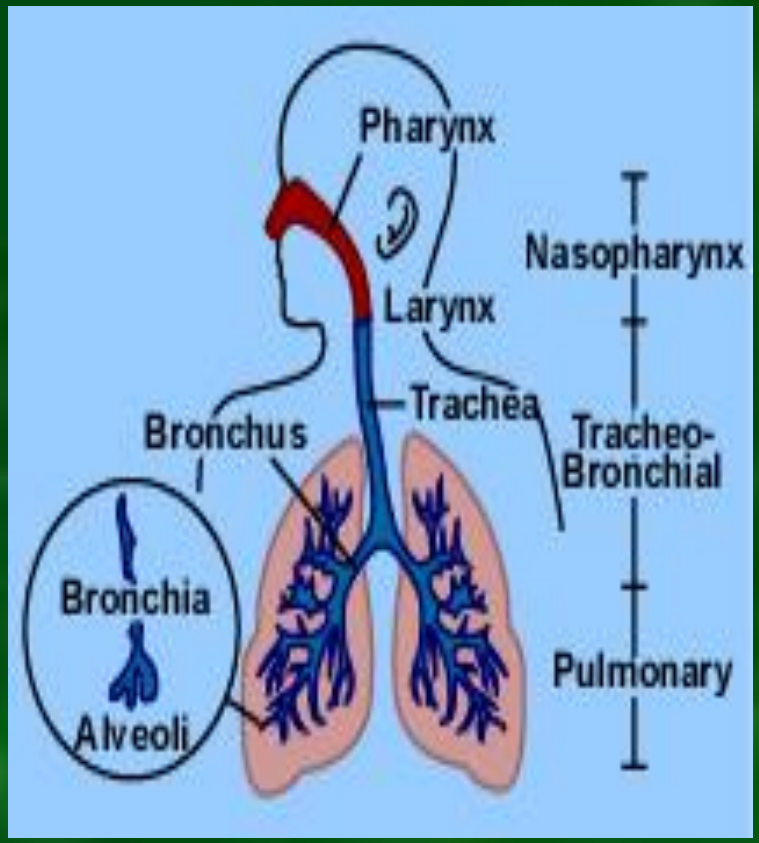
- Exposure to air pollution is associated with numerous effects on human health, including pulmonary, cardiac, vascular, and neurological impairments.
- The health effects vary greatly from person to person. High-risk groups such as the elderly, infants, pregnant women, and sufferers from chronic heart and lung diseases are more susceptible to air pollution.
- Children are at greater risk because they are generally more active outdoors and their lungs are still developing.



- Exposure to air pollution can cause both acute (short-term) and chronic (long-term) health effects.
- **Acute effects** are usually immediate and often reversible when exposure to the pollutant ends. Some acute health effects include eye irritation, headaches, and nausea.
- **Chronic effects** are usually not immediate and tend not to be reversible when exposure to the pollutant ends.
  - Some chronic health effects include decreased lung capacity and lung cancer resulting from long-term exposure to toxic air pollutants.

Effects on Human respiratory system

- Both gaseous and particulate air pollutants can have negative effects on the lungs.
- Solid particles can settle on the walls of the trachea, bronchi, and bronchioles.
- Continuous breathing of polluted air can slow the normal cleansing action of the lungs and result in more particles reaching the lower portions of the lung.
- Damage to the lungs from air pollution can inhibit this process and contribute to the occurrence of respiratory diseases such as bronchitis, emphysema, and cancer.



Dangers of lead and arsenic poisoning

Arsenic poisoning

Nerve damage

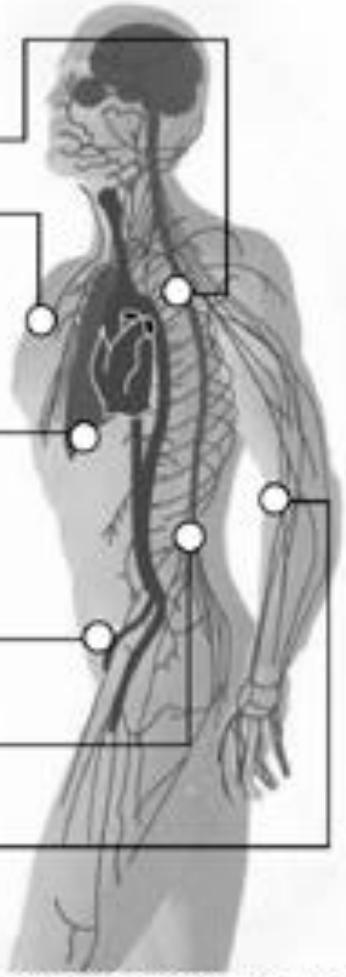
Skin damage:

- Hyperkeratosis (scaling skin)
- Pigment changes

Increased cancer risk:

- Lung
- Bladder
- Kidney and liver cancers

Circulatory problems in skin



Lead poisoning

High levels of lead

- Mental retardation, coma, convulsions and death

Low levels of lead

- Reduced IQ and attention span, impaired growth, reading and learning disabilities, hearing loss and a range of other health and behavioral effects.

Sources: Alliance to End Childhood Lead Poisoning and news wires

The Denver Post

- around 30-40% of cases of asthma and 20-30% of all respiratory disease.
- effect our health in many ways with both short term and long term effect.
- Short term effect are: irritation to nose, eye, throat, bronchitis, headache etc.
- Long term affect are: lung disease, chronic respiratory problem, damage to heart, brain, eyes etc.
- Eye irritation due to NOx, O_3, PAN, particulates.
- Nose and throat due to SO_2, NOx etc.

- Gaseous pollutants like H_2S, SO_2, NO_2 and hydrocarbons cause odor nuisance.
- Irritation of respiration tract caused by SOx, NOx, CO, O_3.
- Increase in mortality.
- High conc. of SO_2, NO_2 and SPM causes bronchitis and asthma.
- CO and NO react with haemoglobin and reduce O_2 carrying capacity of blood.
- Heavy metals like lead can cause poisoning. High conc. cause damage to liver and kidney.

Factors affecting human health

- Nature of the pollutants
- Concentration of the pollutants
- Duration of exposure
- State of health of receptor
- Age group of the receptor



Major air pollutant and their effect

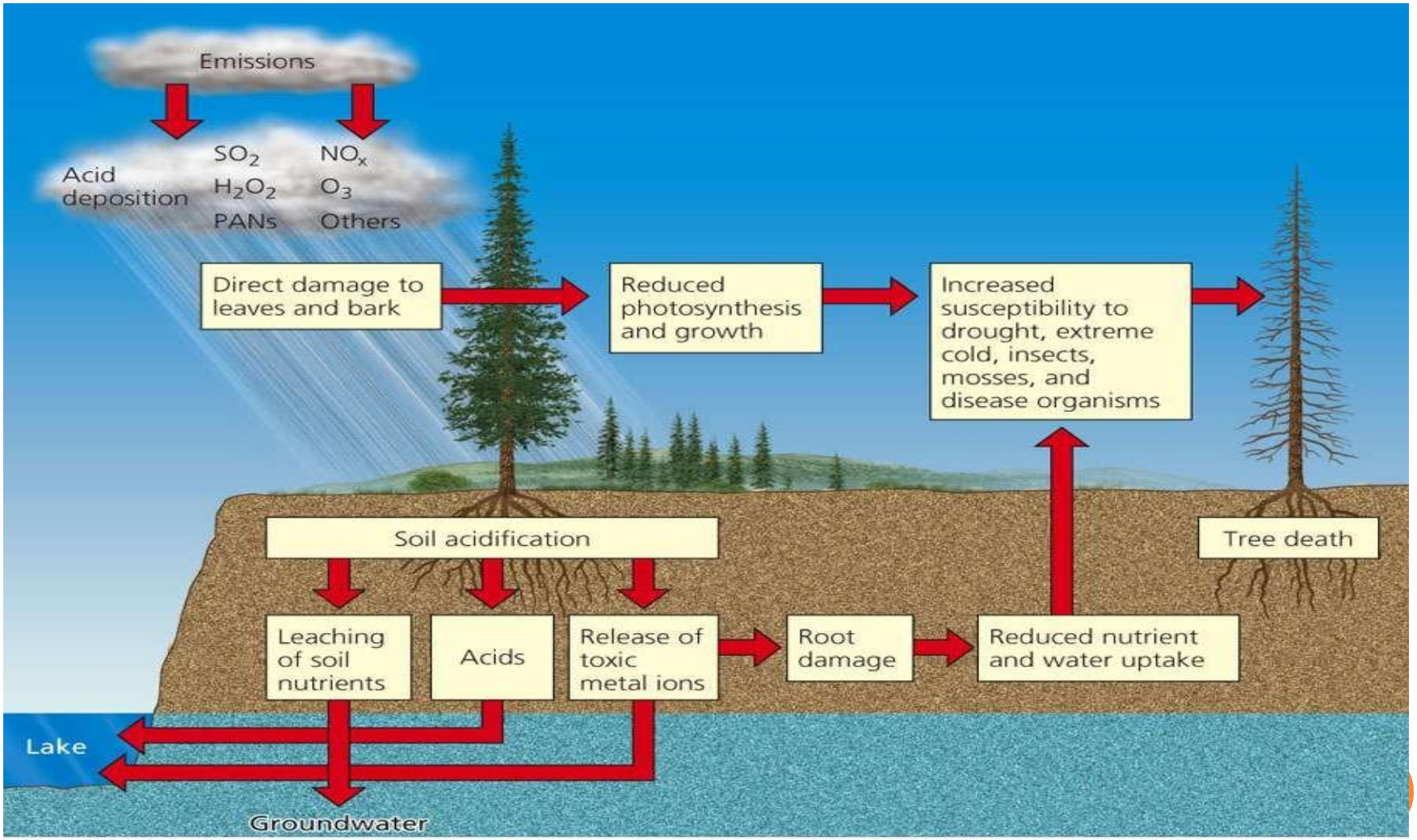


- **Acid Rain** or **Acid Deposition**

a form of precipitation that contains high levels of sulfuric or nitric acids, can contaminate drinking water and vegetation, damage aquatic life, and erode buildings



EFFECTS OF AIR POLLUTION ON PLANT



© Brooks/Cole, Cengage Learning

- Decrease yield in agriculture.
- Suppressed growth of vegetables.
- Leaf injury and damage to young plants.
- Decreased growth rate and increased death rate.



Effects of air pollution on materials

- Corrosion of metals due to SO_2 in presence of oxygen and moisture is converted into H_2SO_4 acid.
- H_2SO_4 acid react with limestone, marble and other building materials to cause deterioration.
- Soiling and eroding of building materials.
- SO_2, O_3, H_2S and aerosols damage protective coating and paints of the surface.
- O_3 and PAN causes cracking of rubber and various electrical insulations.
- Deterioration of art work due to SPM.

Monitoring of air pollution



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Monitoring of air pollution

- The best indicators of air pollution are :

(1)sulphur dioxide.

(2)smoke and suspended particles:

- A known volume of air is filtered through a white filter paper under specified conditions and the stain is measured by photoelectric meter.
- Smoke concentration is estimated and expressed as micrograms/cubic metre of air as an average level over a period of time.

Monitoring of air pollution

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Monitoring of air pollution

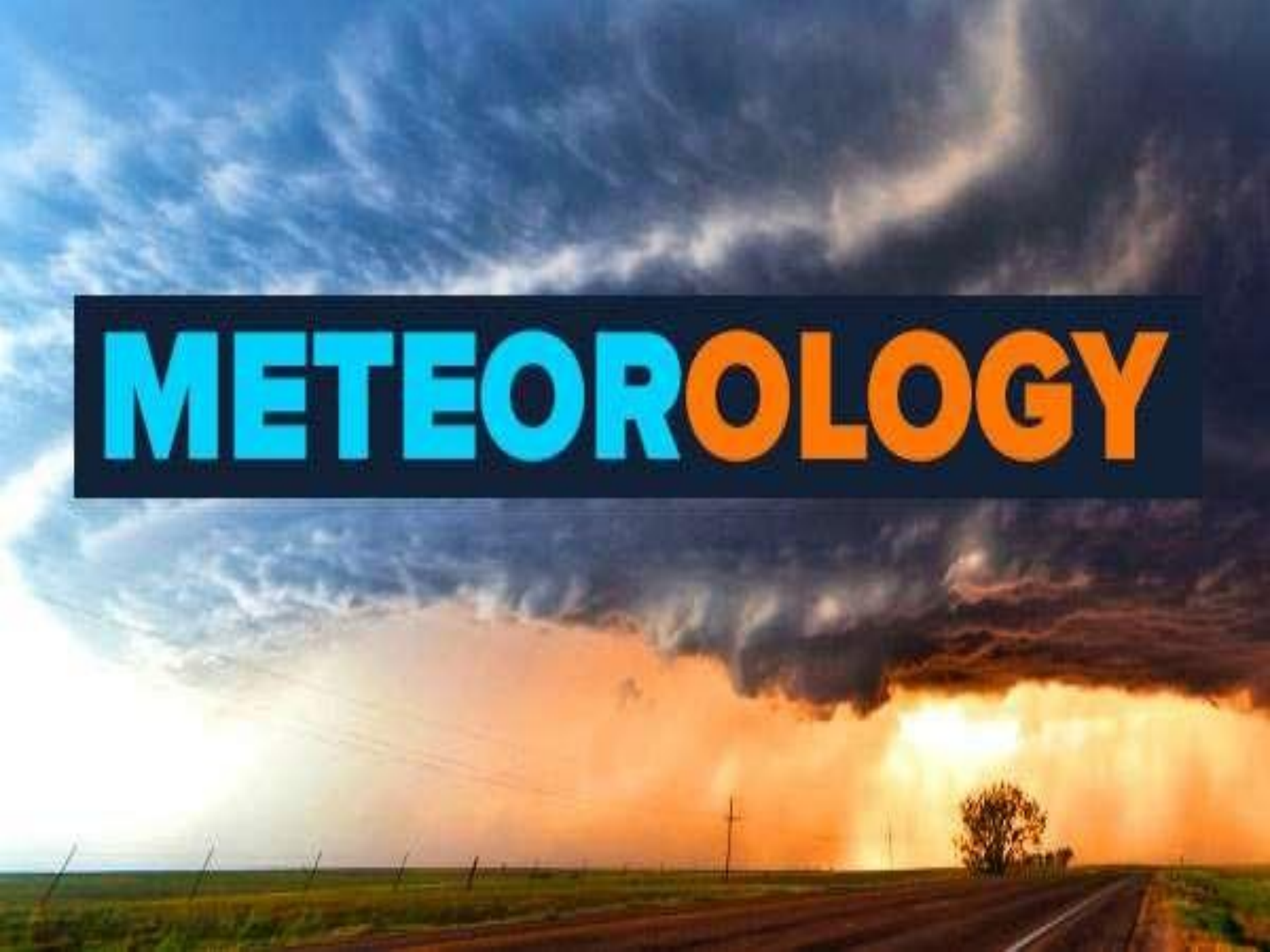
- (c) *Grit and dust measurement :*
- *Deposit gauges collect grit, dust and other* solids. There are analyzed monthly.
- (d) *Coefficient of haze:*
- *A* factor used, particularly in the USA in assessing the amount of smoke or other aerosol in air.

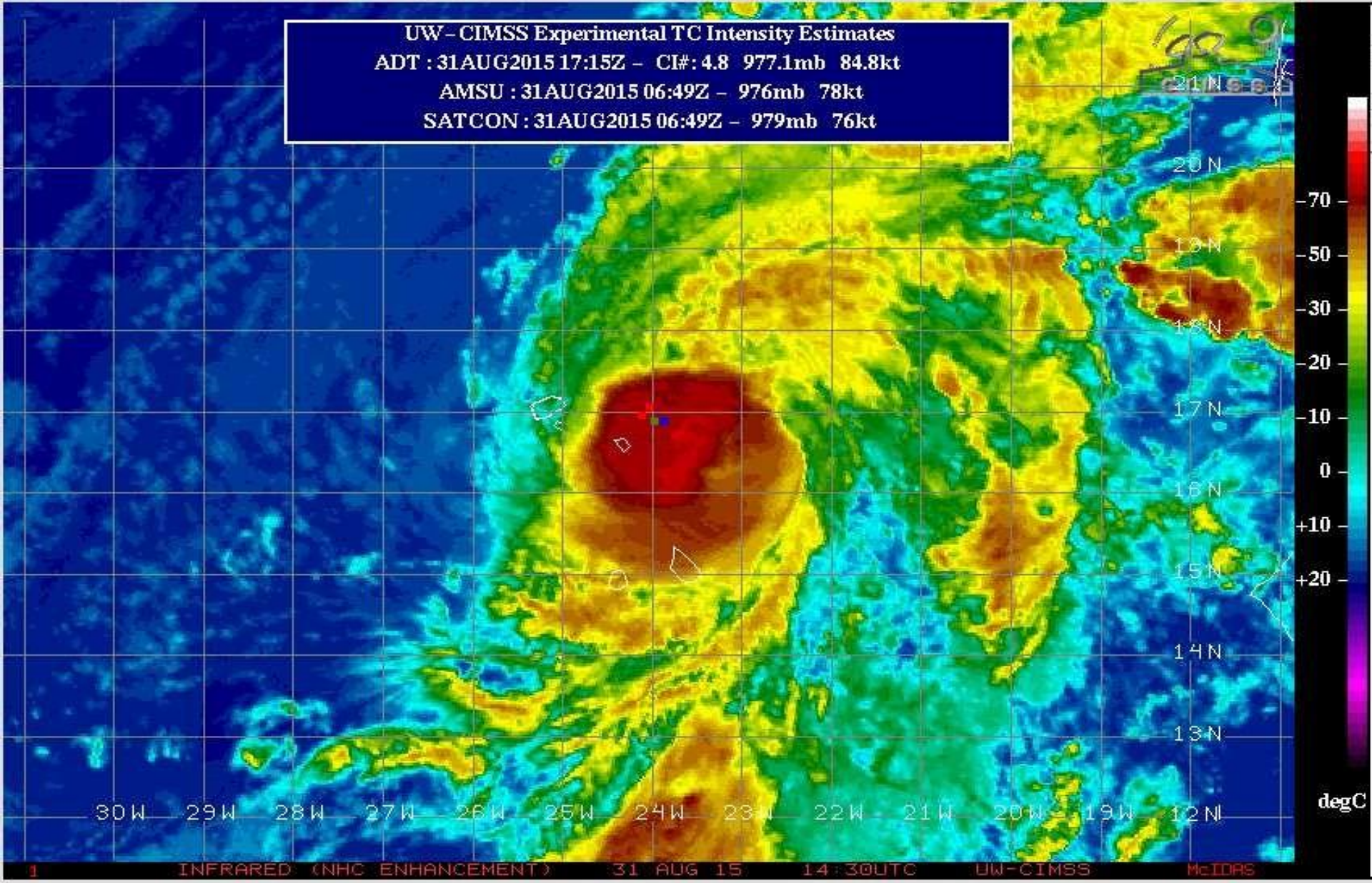
AQI: Air Quality Index

- Indicates whether pollutant levels in air may cause health concerns.
- Ranges from 0 (least concern) to 500 (greatest concern)



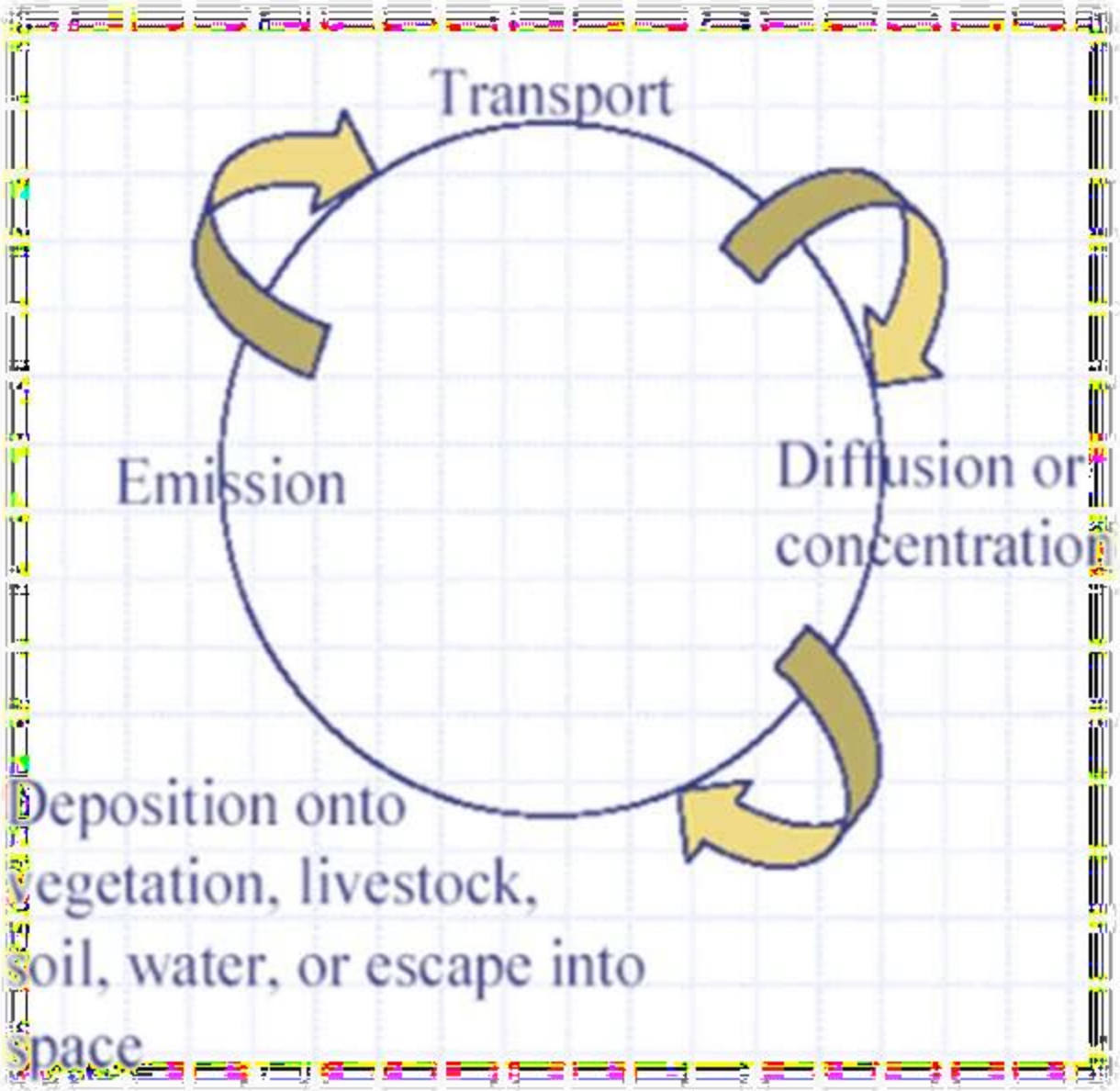
Air Quality	Air Quality Index	Protect Your Health
Good	0-50	No health impacts are expected when air quality is in this range.
Moderate	51-100	Unusually sensitive people should consider limiting prolonged outdoor exertion.
Unhealthy for Sensitive Groups	101-150	Active children and adults, and people with respiratory disease, such as asthma, should limit prolonged outdoor exertion.
Unhealthy	151-200	Active children and adults, and people with respiratory disease, such as asthma, should limit prolonged outdoor exertion, everyone else, especially children should limit prolonged outdoor excertion.
Very Unhealthy (Alert)	201-300	Active children and adults, and people with respiratory disease, such as asthma, should limit prolonged outdoor exertion everyone else, especially children, should limit outdoor exertion.


METEOROLOGY



- 0-1 Hours Previous
- 1-2 Hours Previous
- 2-3 Hours Previous

Air Pollutant Cycle

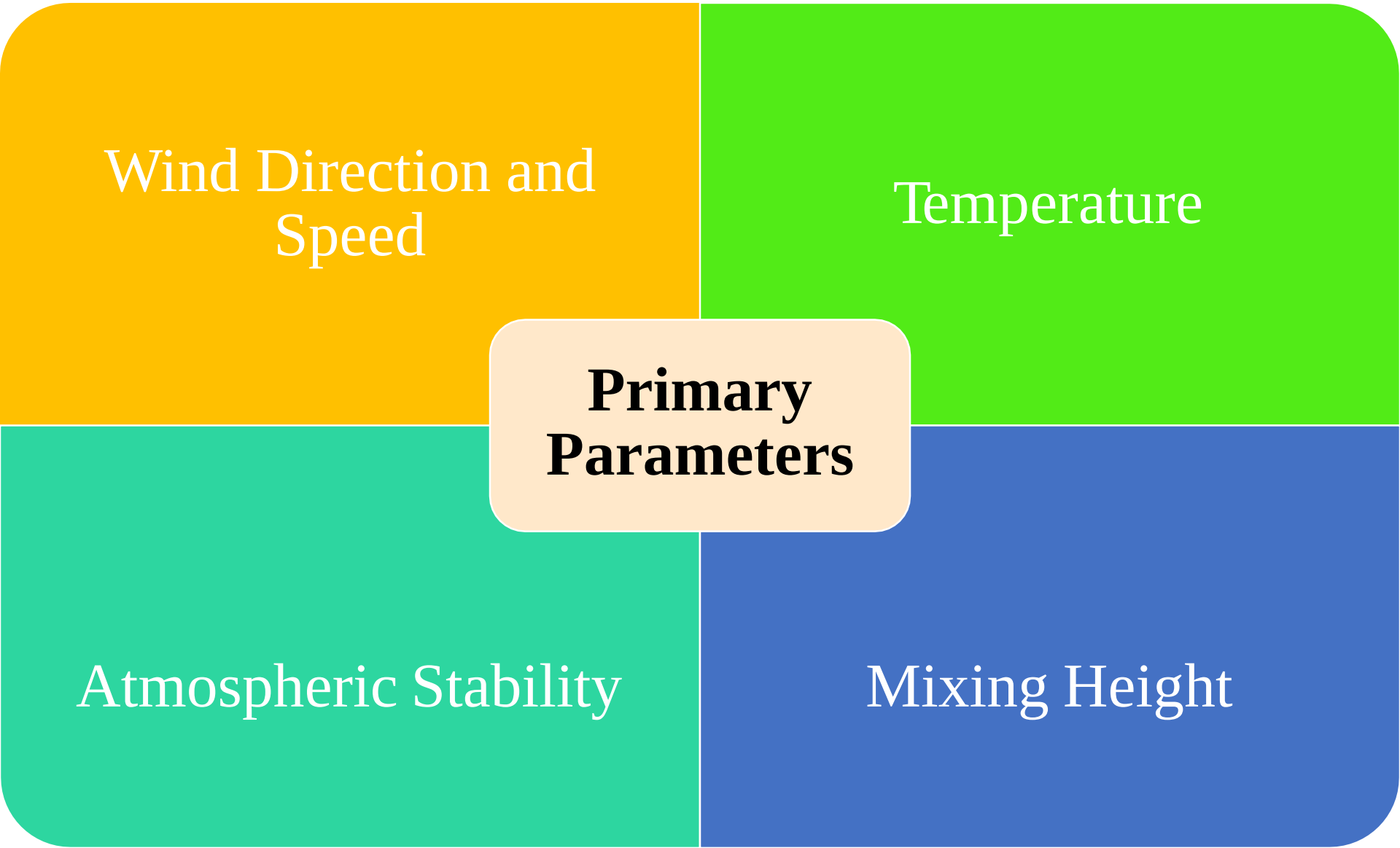


Meteorological Factors Influencing Air Pollution

Primary Parameters

Secondary Parameters

Primary Parameters

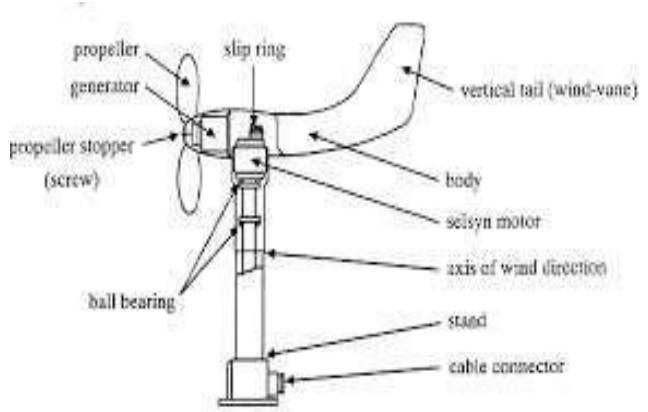


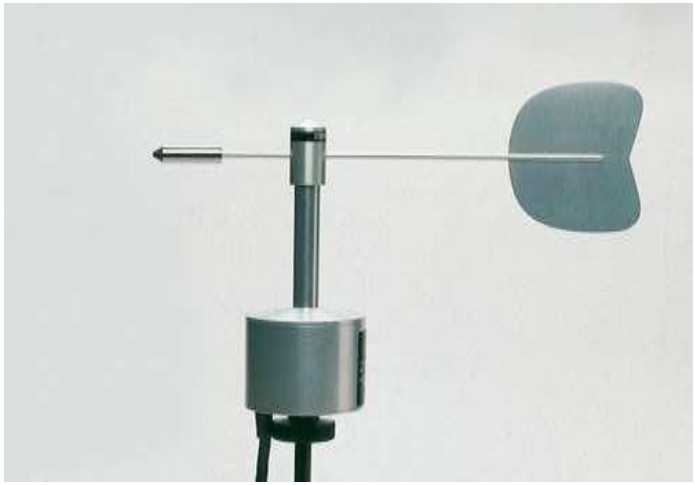
Methods for Measurement of Meteorological Variables

- Wind Direction Recorder
- Wind Speed Recorder
- Temperature Measurement
- Solar Radiation Measurement

Wind Direction Recorder

- Flat Plate vane
- Splayed vane
- Aerofoil vane
- Running average anemograph





Wind Direction Aloft

- Pilot Balloons (Pibals)
- Tetroons
- Kite Balloons
- Radio and Radar
- Smoke trails

Wind Direction Aloft









Wind Speed Recorder



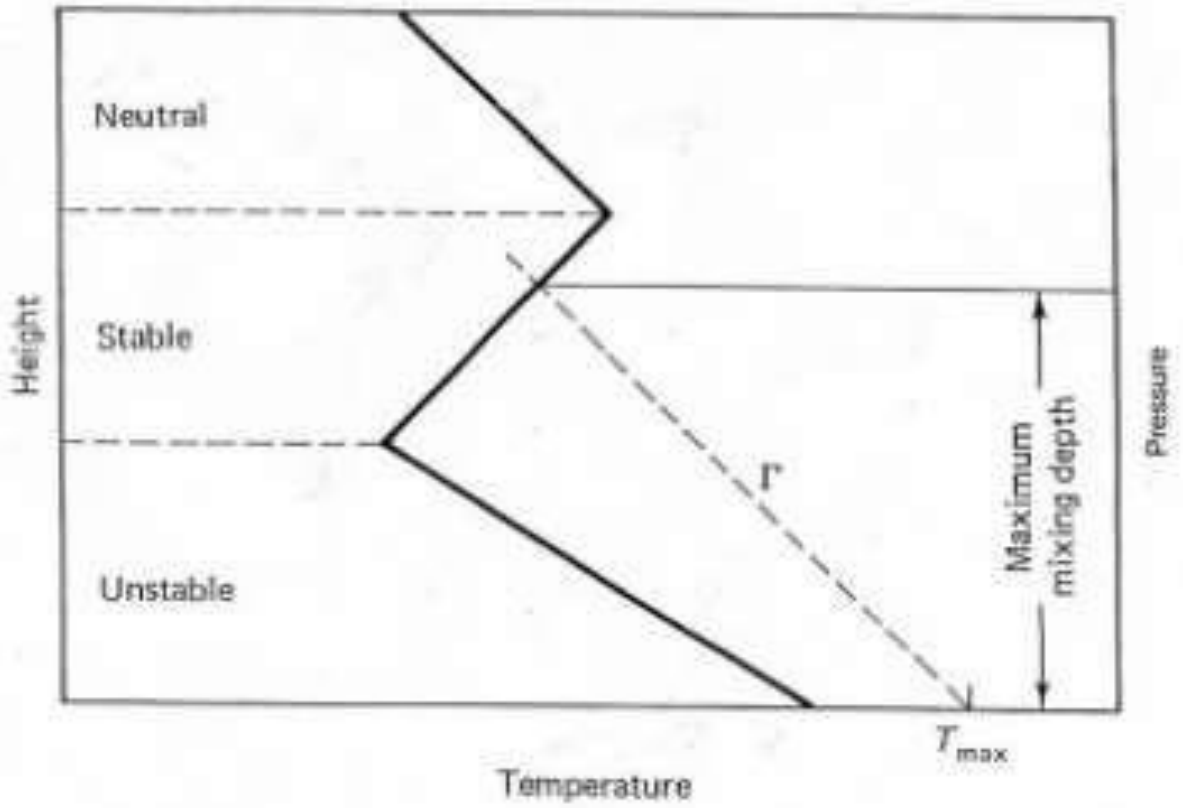


Fog



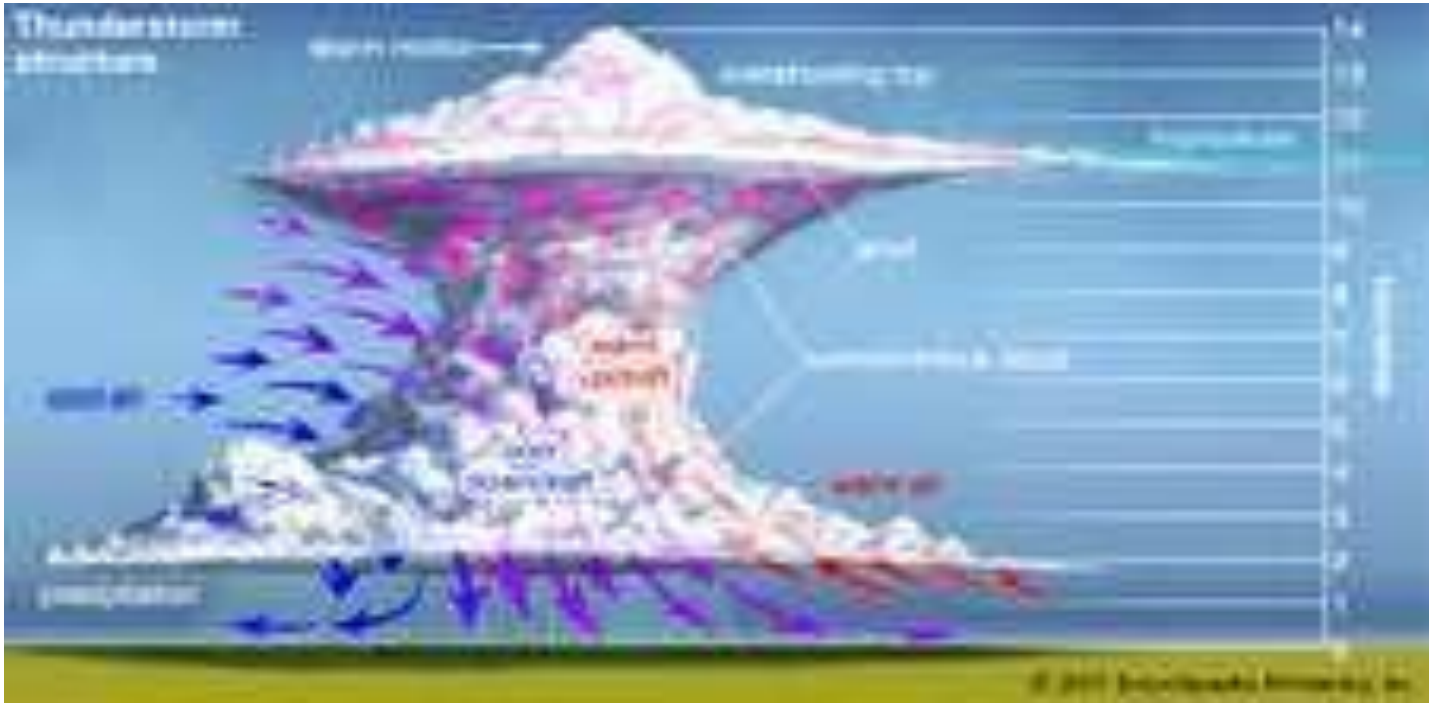
Mixing Height

The **mixing height** is the height of vertical mixing of air and suspended particles above the ground. This height is determined by the observation of the atmospheric temperature profile.

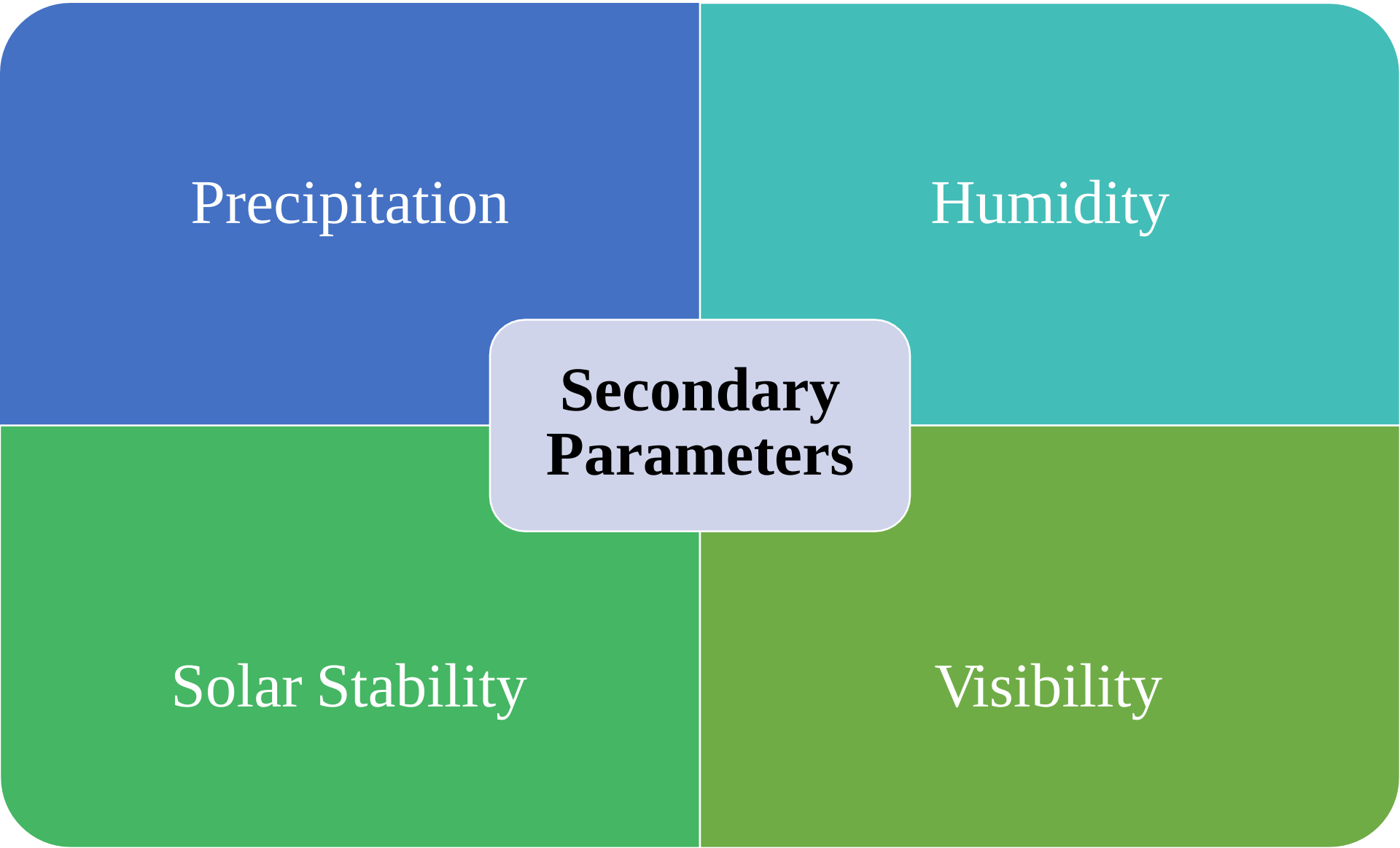


Atmospheric Turbulence

- **Atmospheric turbulence**, small-scale, irregular air motions characterized by winds that vary in speed and direction. **Turbulence** is important because it mixes and churns the **atmosphere** and causes water vapour, smoke, and other substances, as well as energy, to become distributed both vertically and horizontally.



Secondary Parameters



Precipitation



Humidity Measurement
-
Whirling Psychrometer



Humidity Measurement



KETECH Instrument



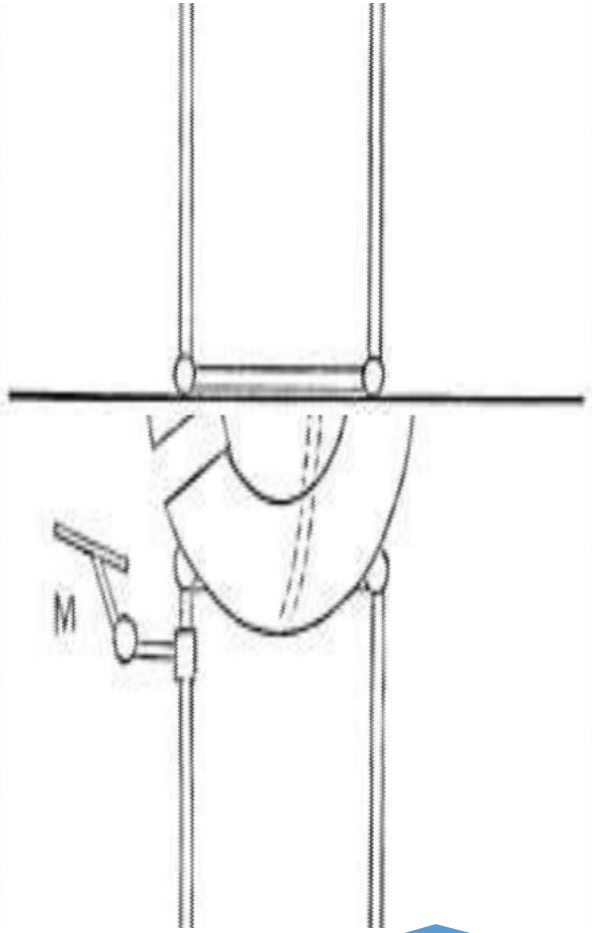
Solar Radiation Measurement



Pyrheliometer



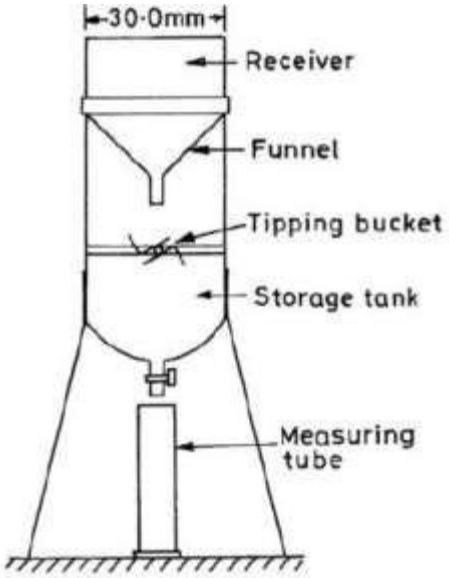
Solarimeter



Actinometer



Anemometer



Rain Gauge



Hygrometer

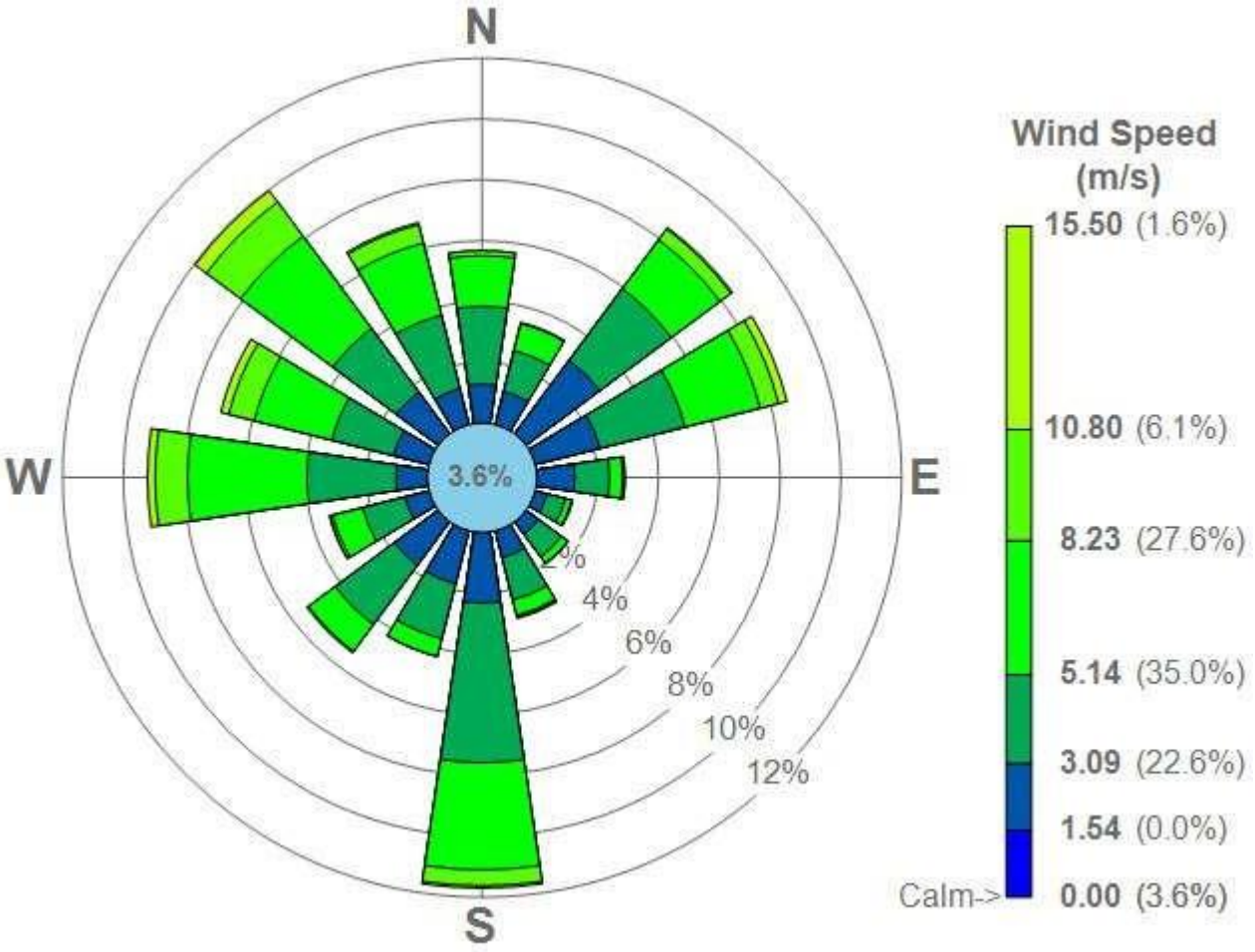


Pyranometer

Visibility

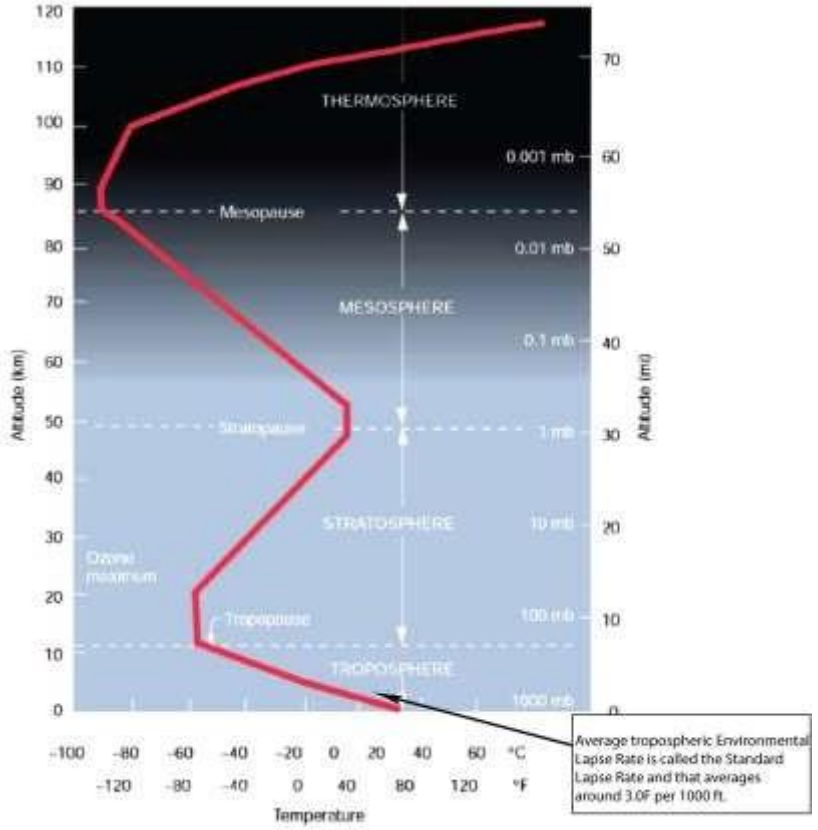


Wind Rose Diagram



Lapse Rate

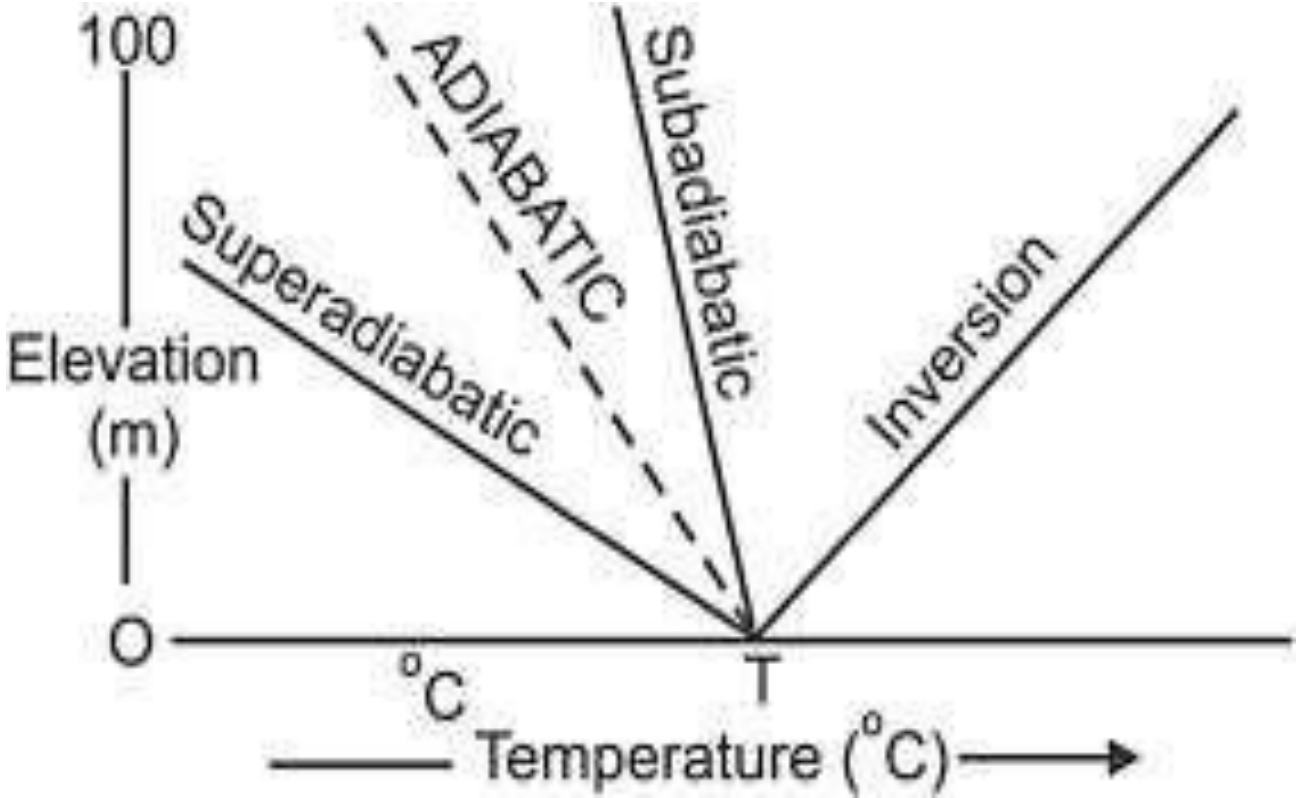
- The **lapse rate** is the **rate** at which an atmospheric variable, normally temperature in Earth's atmosphere, changes with altitude. **Lapse rate** arises from the word **lapse**, in the sense of a gradual change.



Lapse Rate

- **Adiabatic lapse rate**: Change of temperature with a change in altitude of an air parcel without gaining or losing any heat to the environment surrounding the parcel.
- **Dry adiabatic lapse rate**: Assumes a dry parcel of air. Air cools 1°C/100 m rise in altitude (5.4°F/1000 ft).
- **Wet adiabatic lapse rate**: As parcel rises, H_2O condenses and gives off heat, and warms air around it. Parcel cools more slowly as it rises in altitude, ≈6°C/1000 m (≈3°F/1000 ft).
- **Ambient or prevailing lapse rate**: The actual atmospheric temperature change with altitude; not only does water content modify lapse rates, but wind, sunlight on the Earth's surface, and geographical features change actual lapse rates.

- **Superadiabatic**: Ambient lapse rate > adiabatic indicates unstable atmosphere. Vertical motion and mixing processes are enhanced. Dispersion of pollution plume is enhanced.
- **Subadiabatic**: Ambient lapse rate < adiabatic. It indicates stable atmosphere, vertical motion, and mixing are suppressed. Dispersion is suppressed, and contamination is trapped.



Inversion

Temperature actually **increases with altitude** near the **ground** before it begins to **decrease with altitude**. This results in **warm, low-density air** riding on top of **cool high density air**; a very stable air column that traps pollution near the ground

Types of Inversion

- Radiation Inversion

- Subsidence Inversion

- Double Inversion (Sometimes both occur simulatneously)

How they move / transport of air pollutant basics

Plume

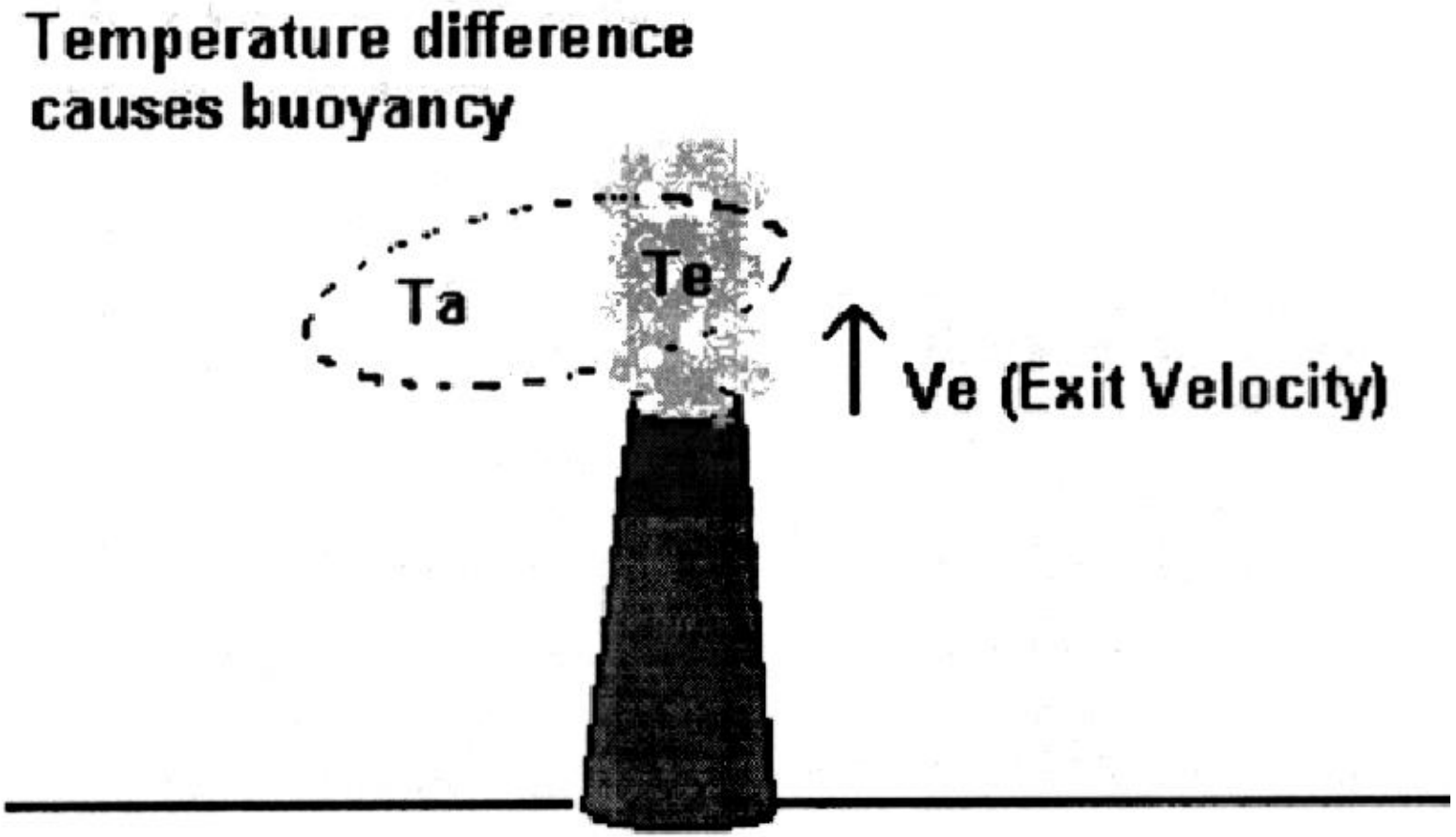
- **Plume** refers to the path and extent in the atmosphere of the gaseous effluents released from a source, usually a stack.



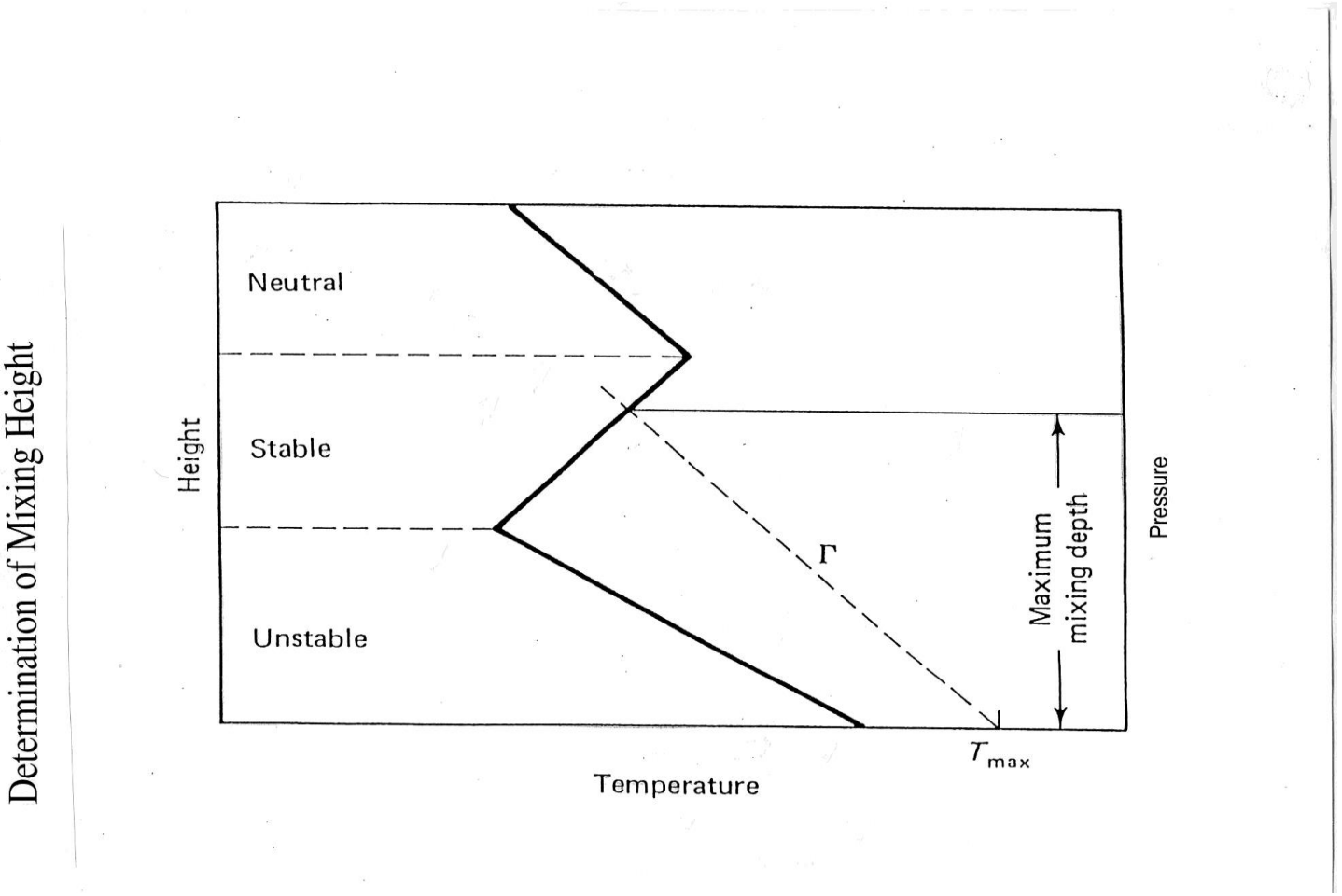
- Mainly follows three laws:
  - Mass transfer: pollutants has mass, and models use this mass
  - Momentum transfer: depends upon movement of pollutants, advection= flow and wind rose diagram helps in understanding it
  - Heat transfer: depends upon lapse rate (change of temperature / increase in height), vertical transport is convective results in heat island

  - In case of stack monitoring all these above stated things are important

Buoyancy = plume rise



- Due to comparison between adiabatic lapse rate (air pollutant / ALR) and environmental lapse rate (ambient air / ELR) results in various stability conditions



Stability affects plume shape / plume behavior



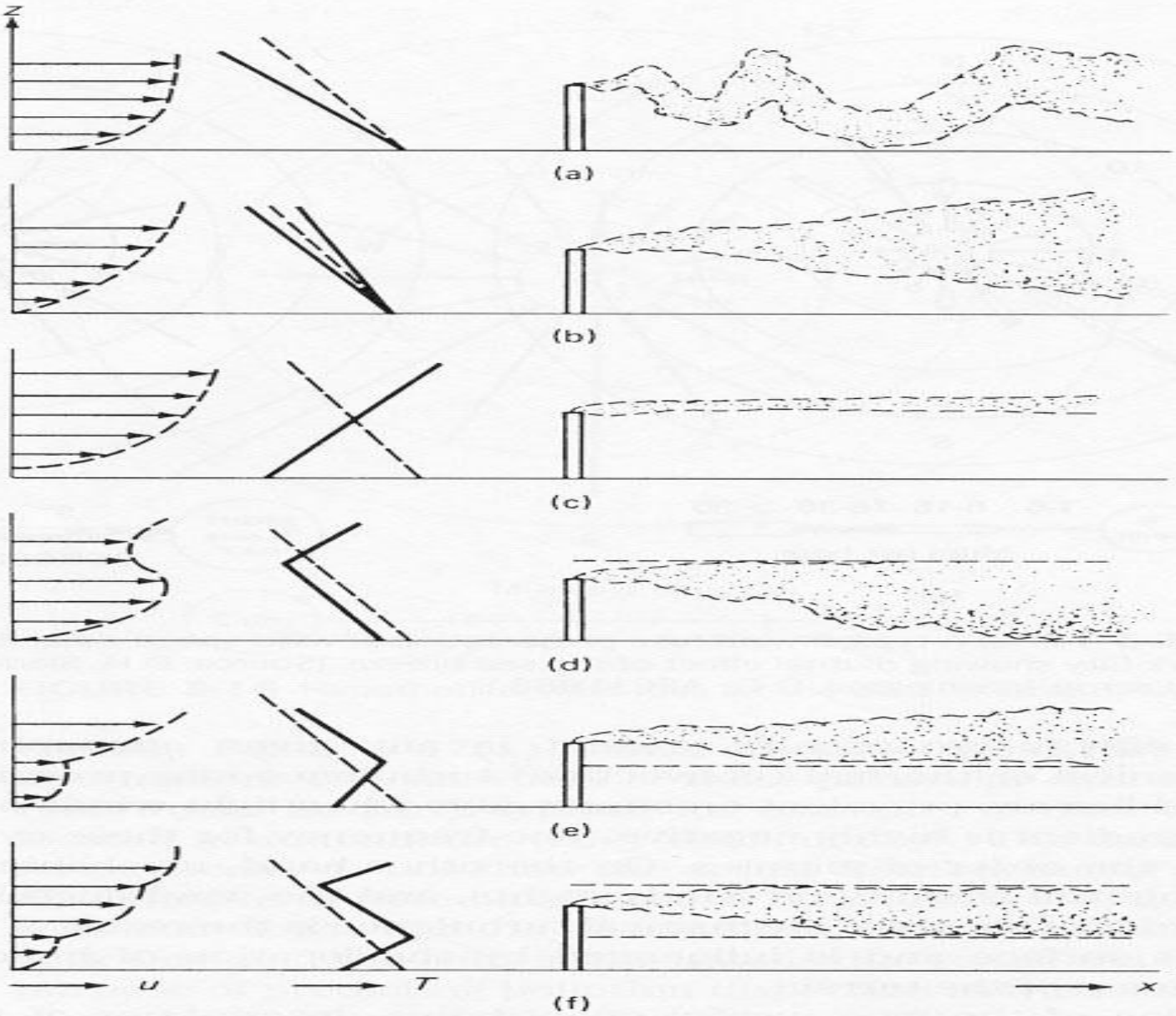
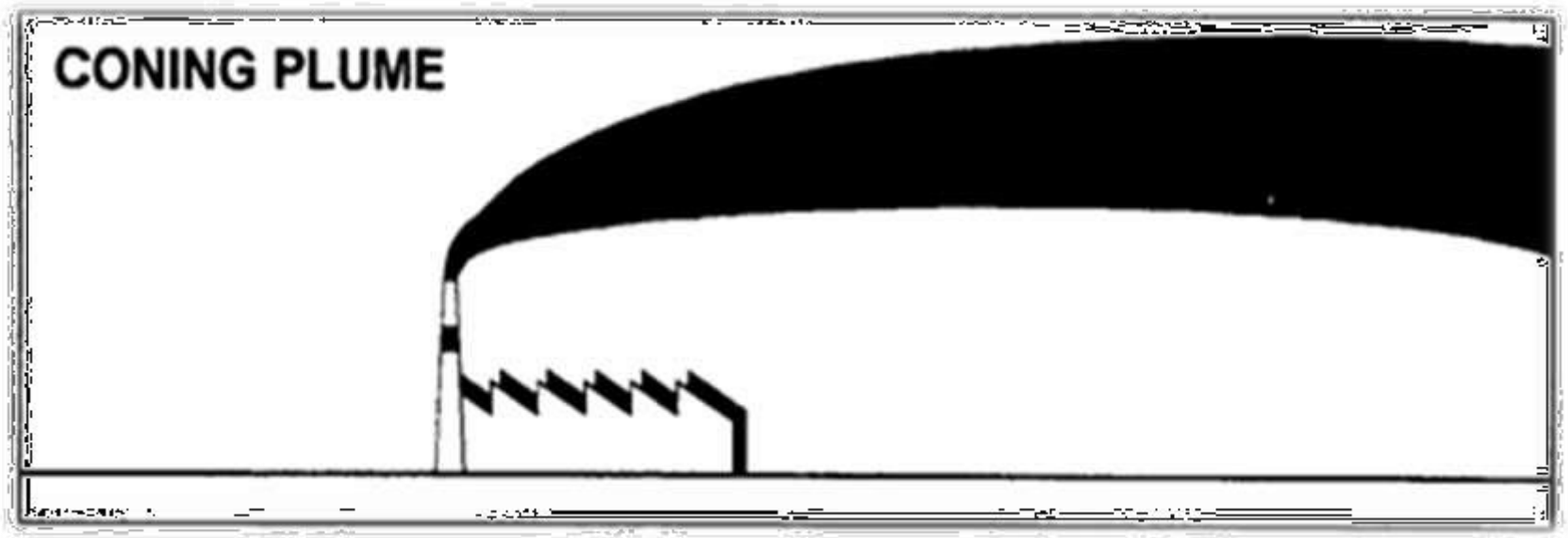
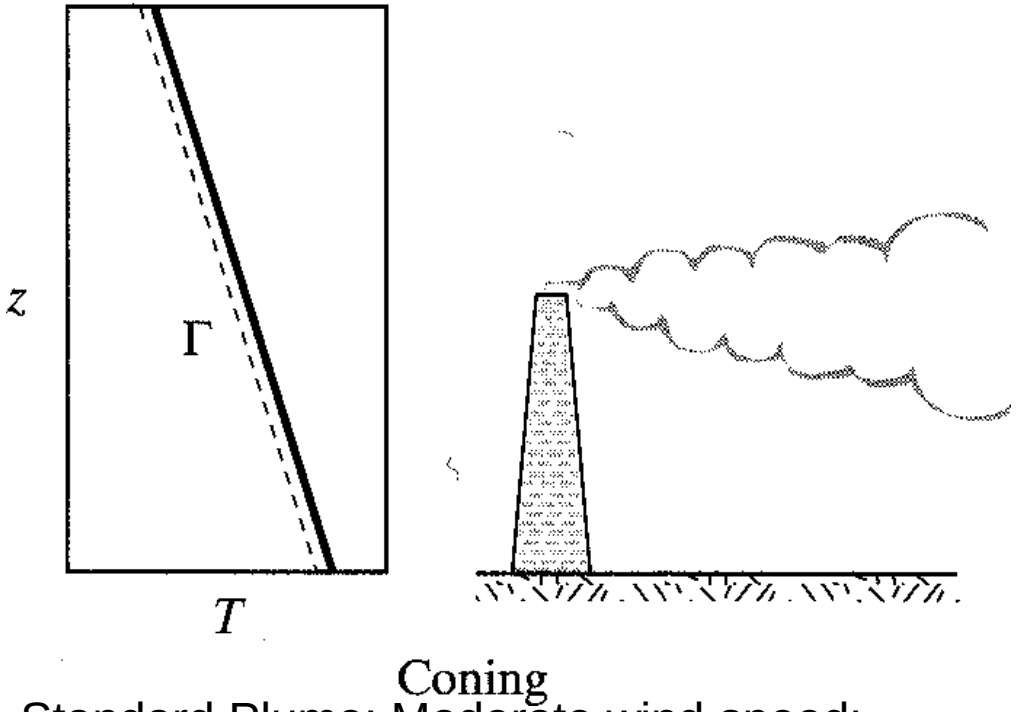


FIGURE 3-18 (a) Typical velocity profile, temperature profile, and plume shape in the x-z coordinate system for various atmospheric conditions. (Dry adiabatic lapse rate - - - ; ambient lapse rate—) (a) Looping, strong instability; (b) coning, near neutral stability; (c) fanning, surface inversion; (d) fumigation, aloft inversion; (e) lofting, inversion below stack; (f) trapping, inversion below and above stack height.

Coning

- Stable with small-scale turbulence
- Associated with overcast moderate to strong winds
- Roughly 10° cone
- Pollutants travel fairly long distances beforereaching ground level in significant amounts
- Occurs in neutral atmospheric conditions





Coning



- Standard Plume: Moderate wind speed:
- Moderate radiation, night time
- Horizontal dispersion at a right angle to the wind is due to turbulence and diffusion, which occurs at the same rate as the vertical dispersion, which is not being opposed nor encouraged by the stability (or lack of it) in the atmosphere.
- Plume spreads equally in the vertical and horizontal as it propagates downstream, forming a coning plume



Example

Z(m)	T(ºC)
10	5.11
202	1.09

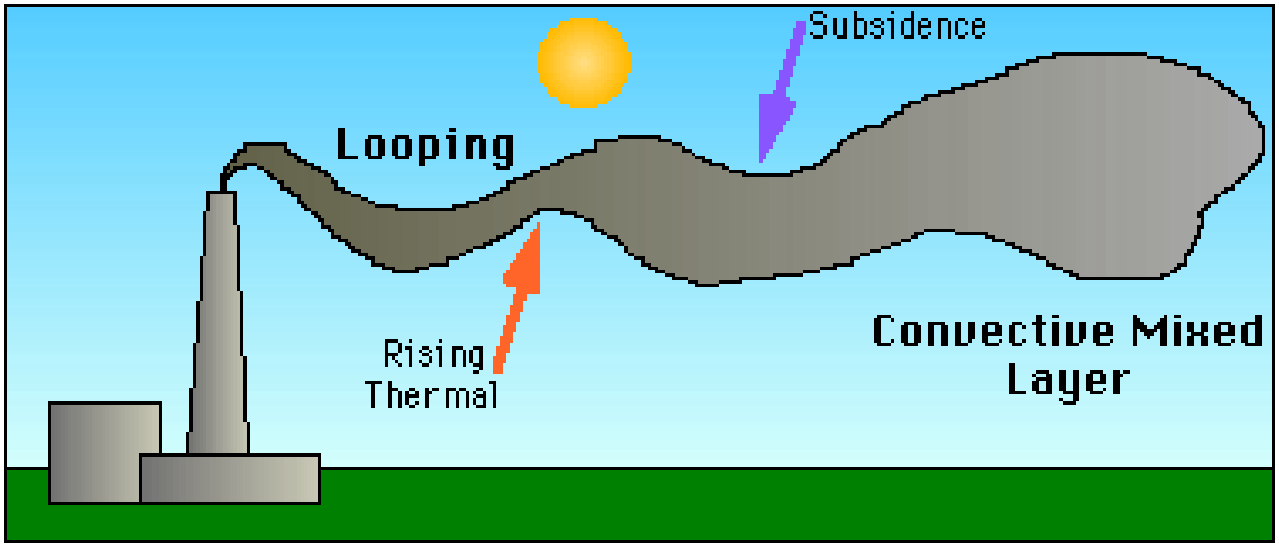
$$\frac{\Delta T}{\Delta z} = \frac{T_2 - T_1}{z_2 - z_1} = \frac{1.09 - 5.11}{202 - 10} = -0.0209 \text{ °C/m}$$

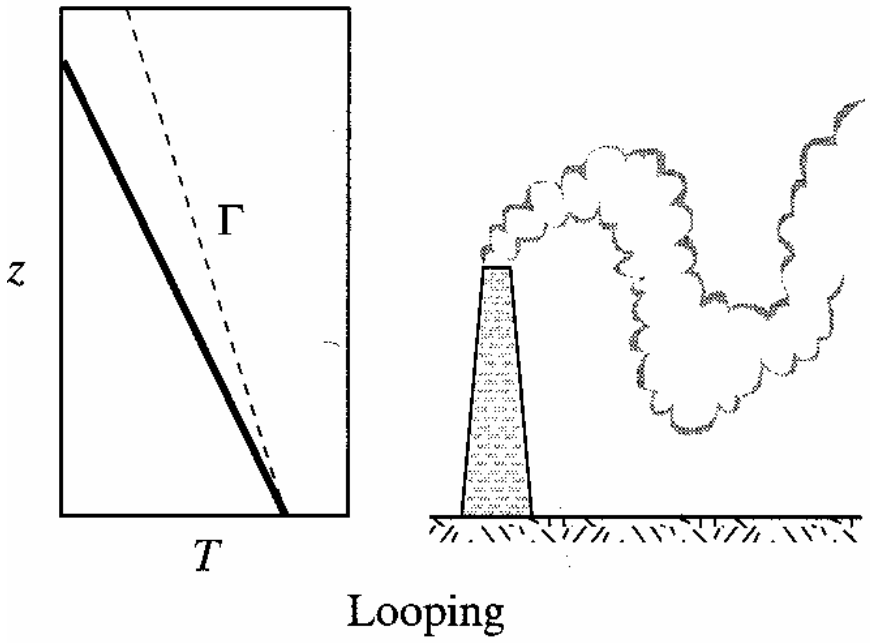
$$= -2.09 \text{ °C/100 m}$$

Since lapse rate is more negative than Γ, (-1.00 ºC/100 m), atmosphere is <u>unstable</u>

Looping

- High degree of convective turbulence
- Superadiabatic lapse rate -- strong instabilities
- Associated with clear daytime conditions accompanied by strong solar heating & light winds
- High probability of high concentrations sporadically at ground level close to stack
- Occurs in unstable atmospheric conditions.





Looping



•In unstable air, the plume will whip up and down as the atmosphere mixes around (whenever an air parcel goes up, there must be air going down someplace else to maintain continuity, and the plume follows these air currents). This gives the plume the appearance that it is looping around.

•Vertical dispersion is very high.

•Less wind speed: Strong & Moderate radiation, day time Mechanical Turbulence is enhanced.

•High probability of high concentrations sporadically at ground level close to stack.

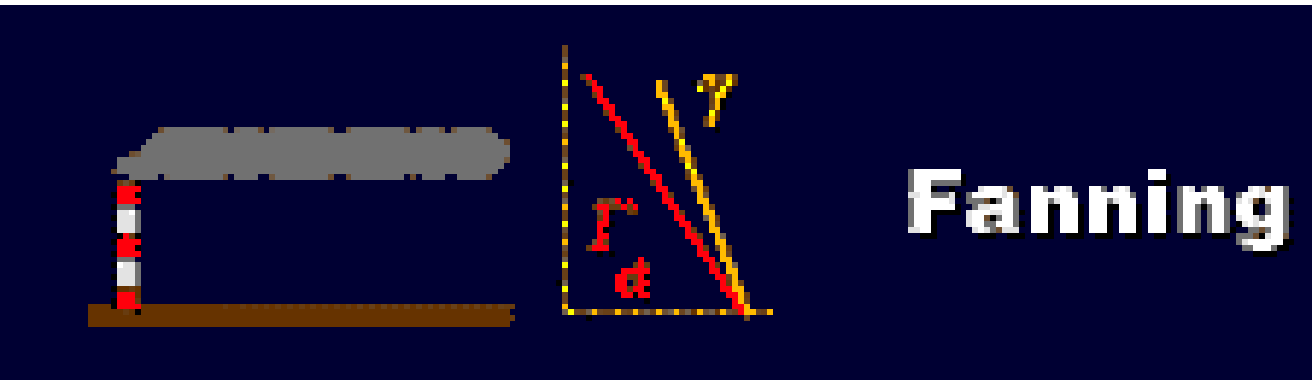
Example

Z(m)	T(ºC)
18	14.03
286	12.56

$$\frac{\Delta T}{\Delta z} = \frac{T_2 - T_1}{z_2 - z_1} = \frac{12.56 - 14.03}{286 - 18} = -0.0055\ °\text{C/m}$$

$$= -0.55\ °\text{C/100m}$$ Suppress Vertical Dispersion

Since lapse rate more positive than Γ, atmosphere is <u>stable</u>



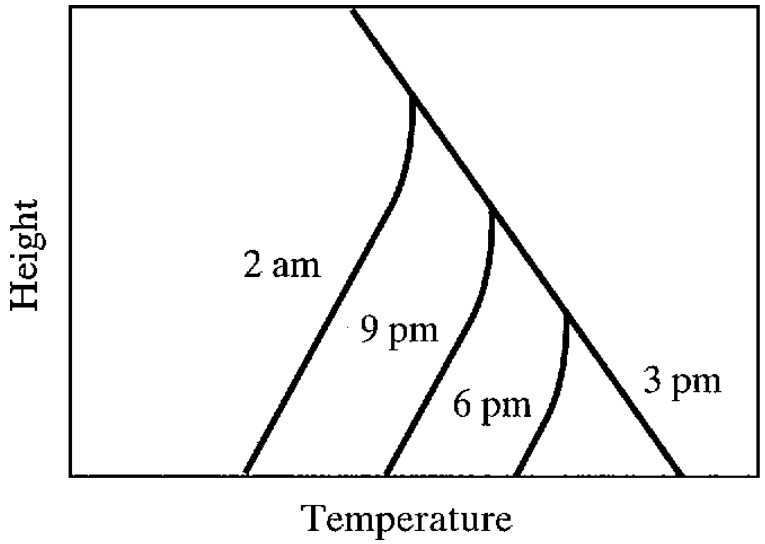


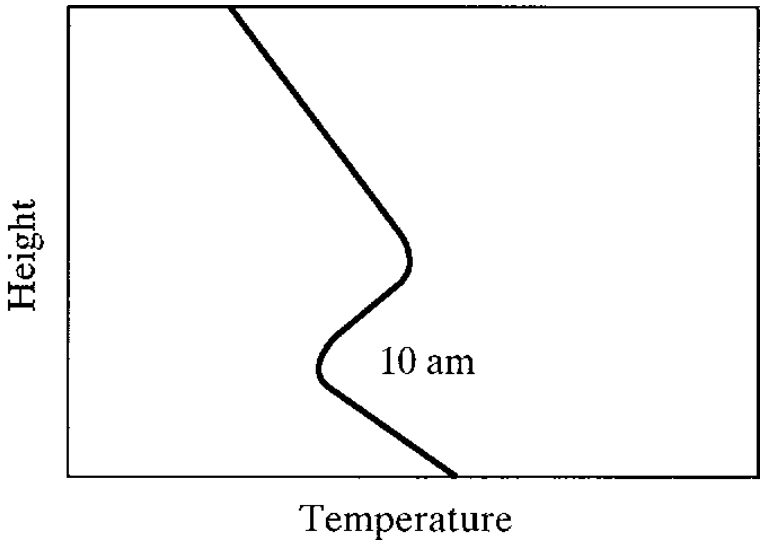
•High wind speed: Night time, High horizontal dispersion, Vertical dispersion is suppresses by stable atmosphere.

•In the vertical, dispersion is suppressed by the stability of the atmosphere, so pollution does not spread toward the ground. This results in very low pollution concentrations at the ground

Temperature Inversions

- Extreme case of stability when lapse rate is actually positive, i.e. temperature increases with altitude
- Resulting temperature inversion prevents nearly all upward mixing

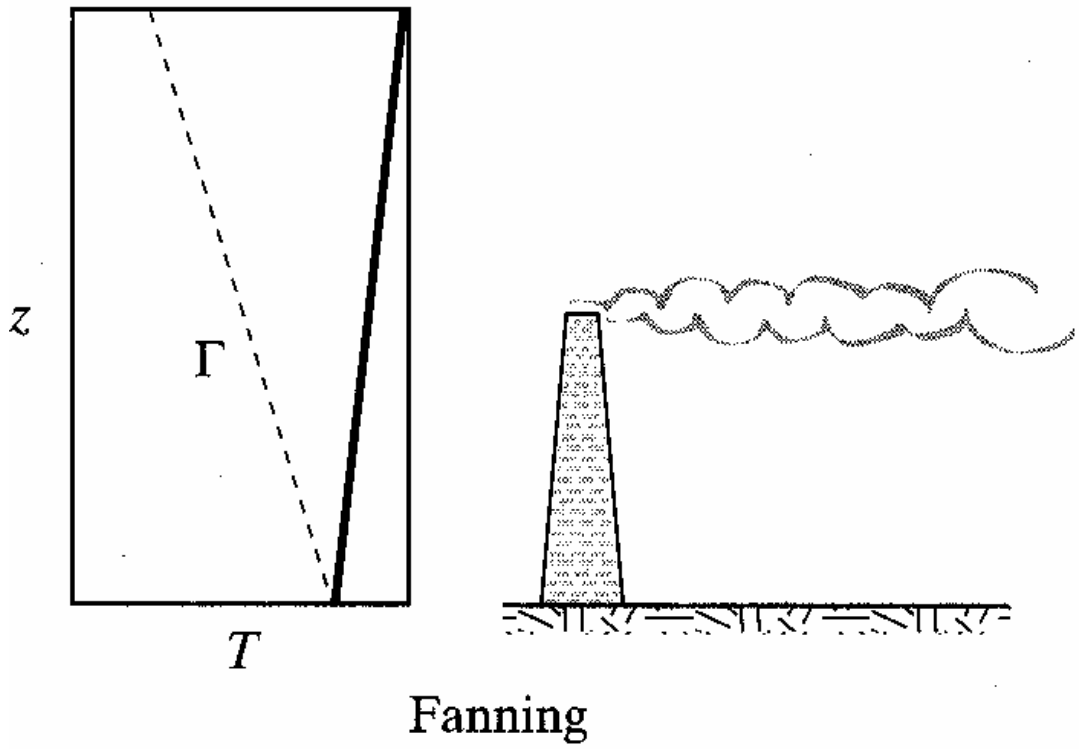




Faning

- Occurs under large negative lapse rate
- Strong inversion at a considerable distance above the stack
- Extremely stable atmosphere
- Little turbulence
- If plume density is similar to air, travels downwind at approximately same elevation





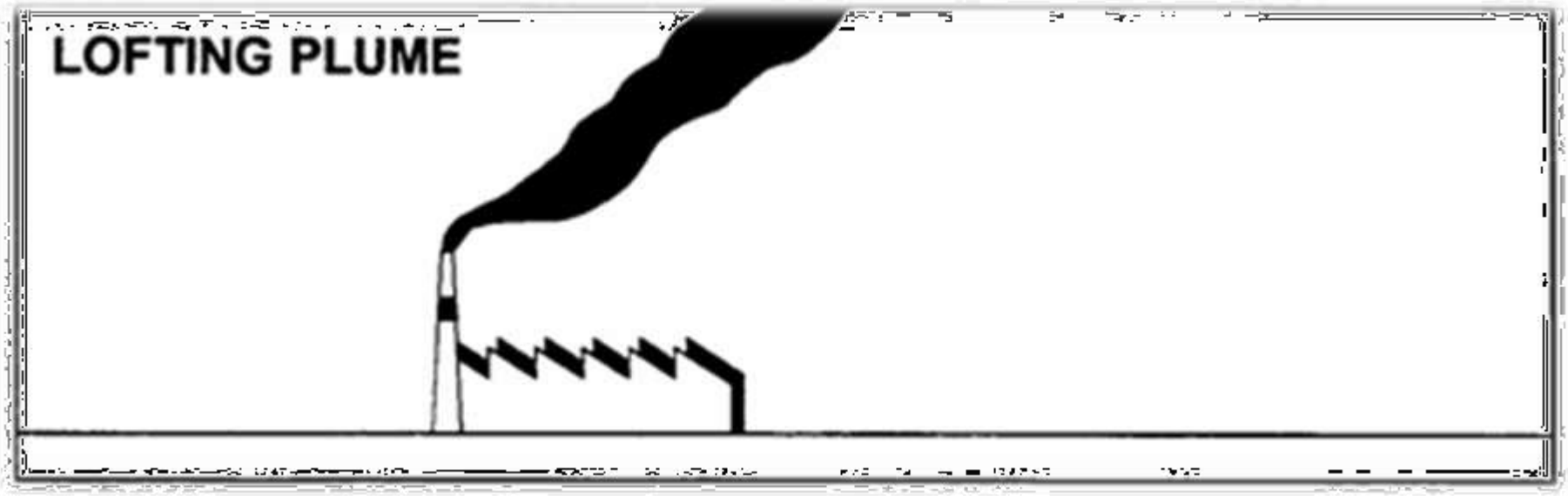
Fanning

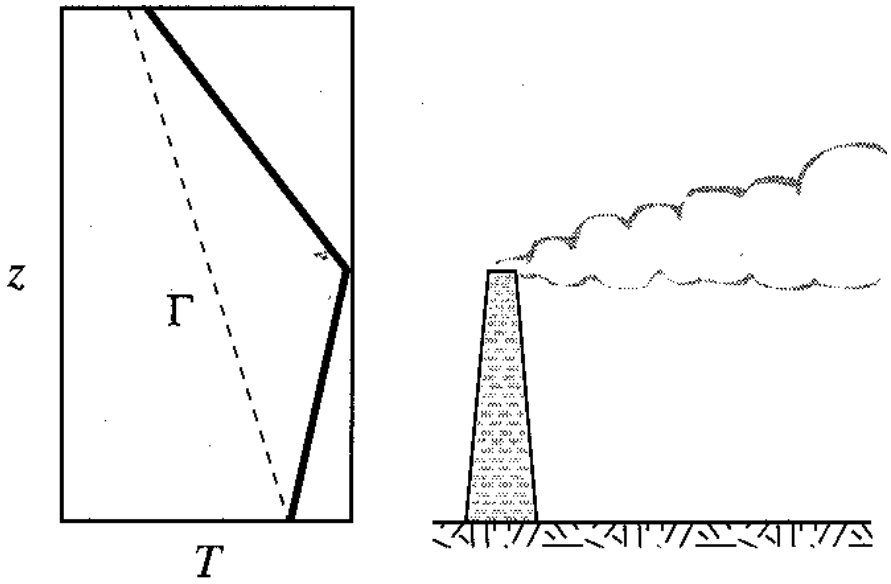


Fanning Plume:
Usually occurs at night, or 1200m-1800m above ground. There is high ground concentration if stack is short or if plume moves through rugged terrain. Occurs in stable inversion atmospheric conditions.

Lofting

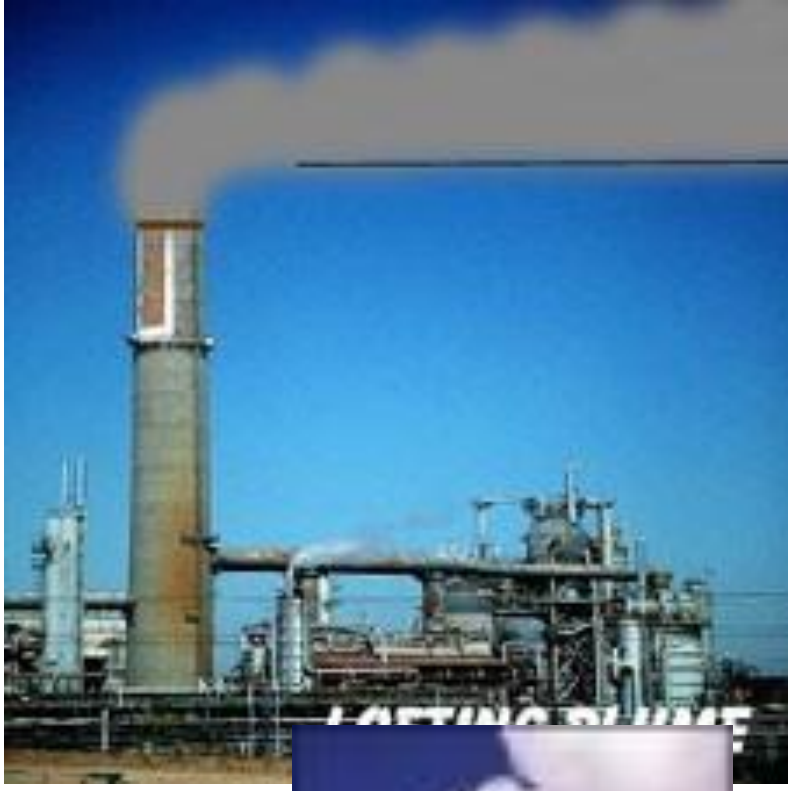
- Favorable in the sense that fewer impacts at ground level.
- Pollutants go up into environment.
- They are created when atmospheric conditions are unstable above the plume and stable below.





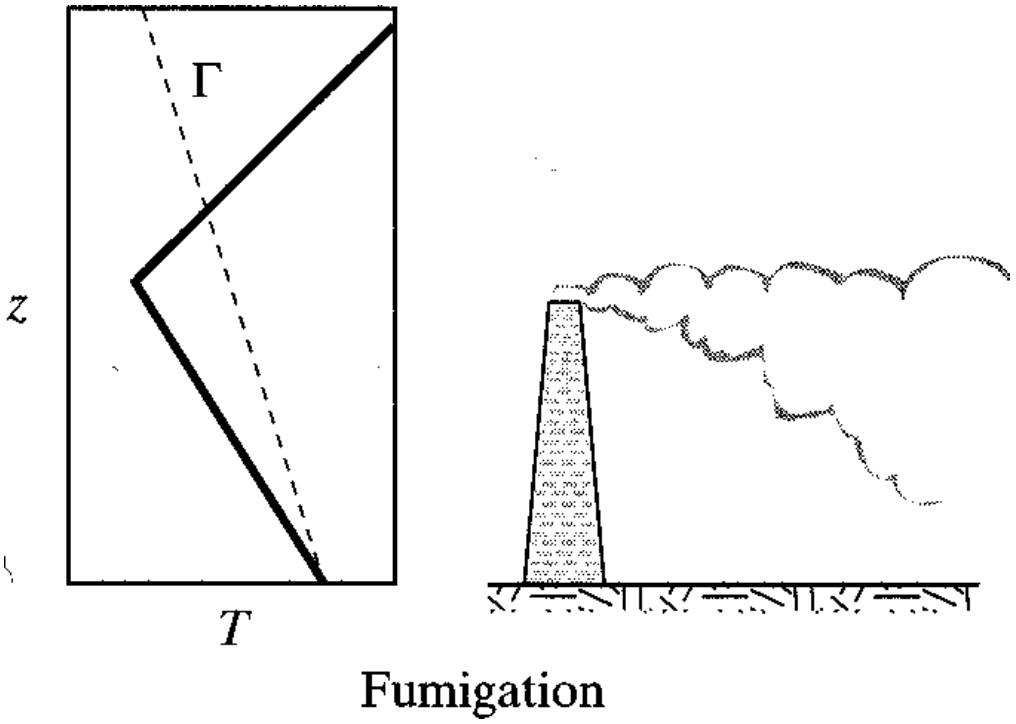
Lofting

(e)





Lofting Plume: favorable in the sense that fewer impacts at ground level. Pollutants go up into environment. They are created when atmospheric conditions are unstable above the plume



Fumigation

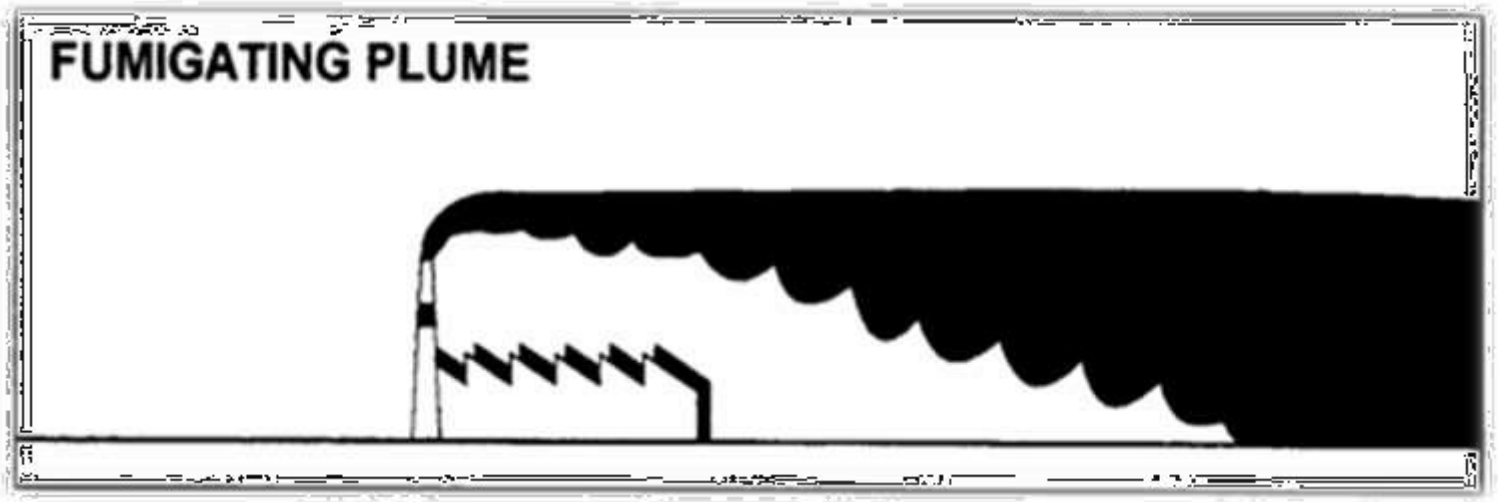




Fumigation:
most dangerous plume: contaminants are all coming down to ground level. They are created when atmospheric conditions are inversion stable above the plume and unstable below. This happens most often after the daylight sun has warmed the atmosphere, which turns a night time fanning plume into fumigation for about a half an hour.

Fumigation

- Most dangerous plume: contaminants are all coming down to ground level.
- They are created when atmospheric conditions are stable above the plume and unstable below.
- This happens most often after the daylight sun has warmed the atmosphere, which turns a night time fanning plume into fumigation for about a half an hour.



Trapping

- When the inversion layer exists above the stack and as well as below the stack, the plume neither goes up nor goes down, rather it gets trapped between these two inversion layers.

- This Plume is not ideal for dispersion of pollutants as it cannot go above a certain height.

Wind profile power law

- The **wind profile power law** is a relationship between the wind speeds at one height, and those at another.
- The wind profile power law relationship is :

$$\frac{u}{u_r} = \left(\frac{z}{z_r}\right)^{\alpha}$$

- Limits
  - The wind profile of the atmospheric boundary layer (surface to around 2000 metres) is generally logarithmic in nature and is best approximated using the log wind profile equation that accounts for surface roughness and atmospheric stability.

Meteorology and Dispersion Modeling
Air Quality and Meteorology

- Primary Metrological Parameter
  –Wind speed, Wind Direction, Atmospheric Stability
- Secondary Metrological Parameter
  –sunlight
  –temperature
  –precipitation and humidity
  –Topography
  –Energy from the sun and earth’s rotation drives atmospheric circulation

Stability

- Dry adiabatic lapse rate – temperature decreases due to lower pressure (ideal gas law)

$$\Gamma = -\frac{dT}{dz} = -1.00\ ^\circ\text{C/100 m} = -5.4\ ^\circ\text{F /1000 ft}$$

- Ambient (actual) lapse rate

  $< \Gamma$ (temperature falls faster) unstable or superadiabatic

  $> \Gamma$ (temperature falls slower) stable or subadiabatic

  $= \Gamma$ (same rate) neutral

Example

Z(m)	T(ºC)
2	-3.05
318	-6.21

$$\frac{\Delta T}{\Delta z} = \frac{T_2 - T_1}{z_2 - z_1} = \frac{-6.21 - (-3.05)}{318 - 2} = \frac{-0.0100\ ^\circ\text{C/m}}{}$$

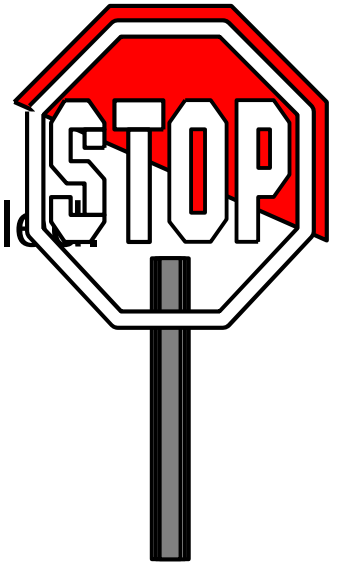
$$= -1.00\ ^\circ\text{C/100 m}$$

Since lapse rate = Γ, atmosphere is neutral

Air pollution control:

Principles of controls, source control

Air pollution control



- Cannot be fully prevented but can be controlled.

1. Preventative measures
2. Control measures using equipments.

Preventative measures (source control)

- Selection of suitable fuel. (Low sulphur coal in power plant, using of CNG)
- Modification in industrial process.
- Selection of suitable site and zoning for industrial unit.

Control measures

- When source control not possible some measures taken to prevent pollution.
- Collecting pollutants by using equipments.
- Destroying the pollutants by thermal or catalytic combustion.
- Changing the pollutants to less toxic form.
- By releasing the pollutants through tall chimneys for greater dispersion.



Control and prevention of air pollution

- WHO has recommended the following procedures for the prevention and control of air pollution:

APPROACHES TO REDUCE INDOOR AIR POLLUTION

1. Eliminate or control the sources of pollution:

- Improved stoves
- Clean fuels (kerosene, gas)
- Venting stoves for cooking and heating
- Regular maintenance of cooking, heating and cooling systems

APPROACHES TO REDUCE INDOOR AIR POLLUTION

2. Ventilation – building design

- Dilute and remove pollutants through ventilation with outdoor air



APPROACHES TO REDUCE INDOOR AIR POLLUTION

3. Air cleaning – NOT air fresheners!

- Air filters and ionizers may remove some airborne particles
- Gas adsorbing material is used to remove gaseous contaminants

APPROACHES TO REDUCE INDOOR AIR POLLUTION

EDUCATION AND ADVOCACY

Education of:

- Children
- Family and community
- Health care providers



APPROACHES TO REDUCE INDOOR AIR POLLUTION

Environment policymaking:

- Framework Convention on Tobacco Control
- Clean indoor air regulations
- Community actions

Research

Control of outdoor air pollution

(a) *Containment:*

- That is, prevention of escape of toxic substances into the ambient air.
- Containment can be achieved by a variety of engineering methods such as enclosure, ventilation and air cleaning.
- A major contribution in this field is the development of "**arresters"** for the removal of contaminants.

Control of outdoor air pollution

- **(b) *Replacement:***
- *That is, replacing a* technological process causing air pollution, by a new process that does not.
- Increased use of electricity, natural gas and central heating in place of coal have greatly helped in smoke reduction.
- There is a move now to reduce lead in petrol which is a cumulative poison.

Control of outdoor air pollution

- (**c**) ***Dilution:***
- *Dilution is* valid so long as it is within the self- cleaning capacity of the environment.
- some air pollutants are readily removed by vegetation.
- The establishment of "green belts“ between industrial and residential areas is an attempt at dilution.
- The capacity for dilution is, however, limited and trouble occurs when the atmosphere is overburdened with pollutants.

green belts



Control of outdoor air pollution

d) *Legislation:*

- e.g., Clean Air Acts, legislation covers such matters as:
  - height of chimneys .
  - powers to local authorities to carry out investigations, research and education concerning air pollution
  - creation of smokeless zones and enforcement of standard for ambient air quality.

OZONE HOLE

- The term ozone hole refers to annual thinning of the ozone layer observed yearly over the South Pole areas,
- whereby the amount of ozone in the atmosphere drastically decreases in local spring
- .Since the beginning of the eighties one notices that this hole is getting deeper and that the covered surface is getting larger.

PREVENTATION BY LAWS

- Various laws has been established for the menace of air pollution.
- Air (Prevention & control of pollution) Act,1981.
- Air (Prevention & control of pollution) Amendment Act,1987.
- Motor vehicle Act, 1988.
- Air (Prevention & control of pollution) Union Territories Rules, 1983.
- Environment Protection Act, 1986.

PARTICULATE CONTROLL TECHNOLOGY:

- Dry Control Devices:

1. Gravity Settling Chamber
2. Cyclone Separator
3. Electrostatic Preceptor
4. Fabric Filter

- Wet Controls Devices

1. Cyclone Scrubber
2. Venturi Scrubber
3. Tray Scrubber (Spray Tower)

- Gaseous Control Technology

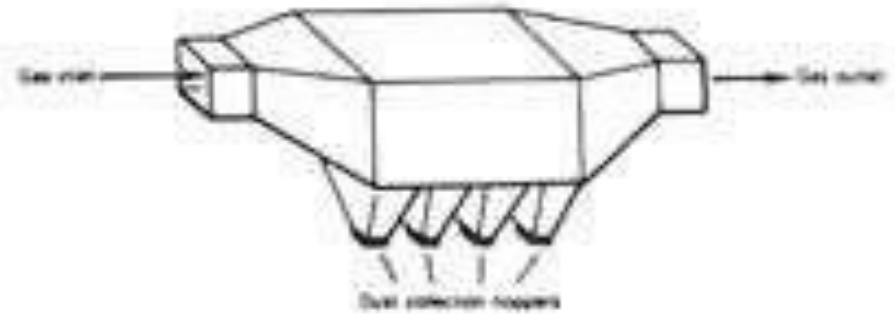
1. Adsorption
2. Absorption
3. Combustion



Dry Control Devices

Air pollution controlling equipments

- Gravitational settling chamber



Settling Chamber

- Used to remove particles with size greater than 50 µm.
- Velocity of flue gas reduced in large chamber.
- Particles settle under gravitational force.

Advantages

- Low initial cost.
- Easy to design.
- Low pressure drop.
- Low maintenance cost.
- Dry and continuous disposal of solid particulates.

Disadvantages

- Require large space.
- Less collection efficiency.
- Only larger size particles can be collected.



Cyclone separator

- Centrifugal force is utilized to separate the particulate matter.
- It can remove 10 to 50 μm particle size.
- Used mostly in industries.



Advantages

- Low initial cost.
- Require less floor area.
- Simple construction and maintenance.
- Can handle large volume of gas at high temp.

Disadvantages

- Requires large head room.
- Less efficiency for smaller particles (<10μm).
- Sensitive to variable dust load and flow rate.

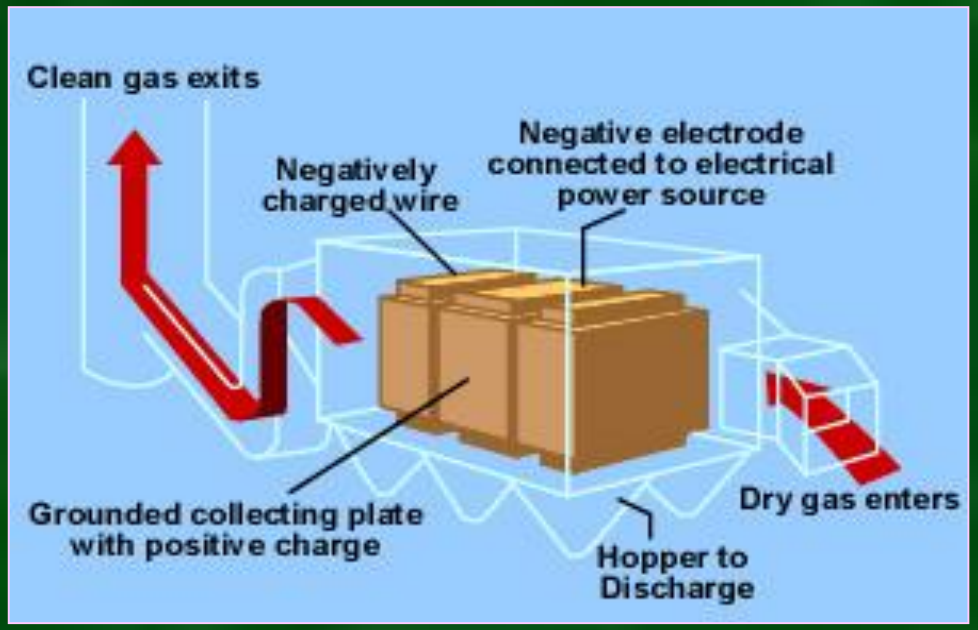
Electrostatic precipitators

- Works on the principle of electrical charging of particulate Matter (-ve) and collecting it in a +ve charged surface.
- 99% efficiency.
- Can remove particle size range of 0.1 μm to 1 μm.



Electrostatic Precipitators (ESPs)

- An ESP is a particle control device that uses electrical forces to move the particles out of the flowing gas stream and onto collector plates.
- The ESP places electrical charges on the particles, causing them to be attracted to oppositely charged metal plates located in the precipitator.



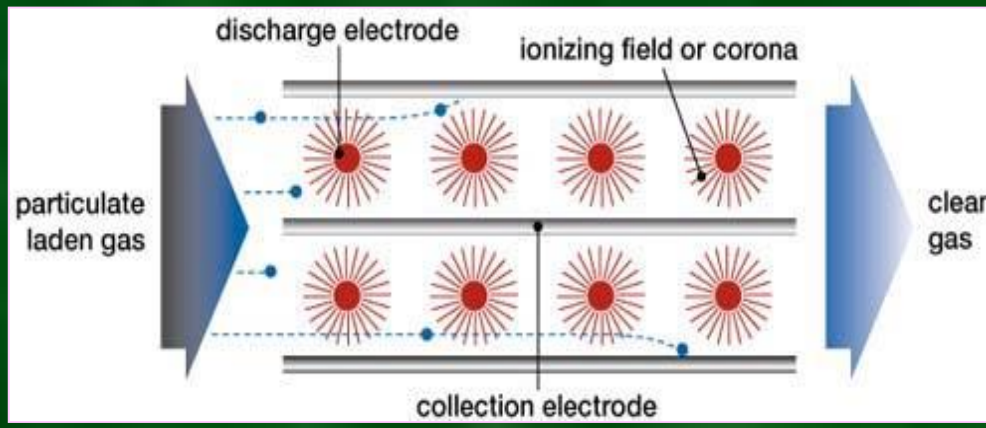
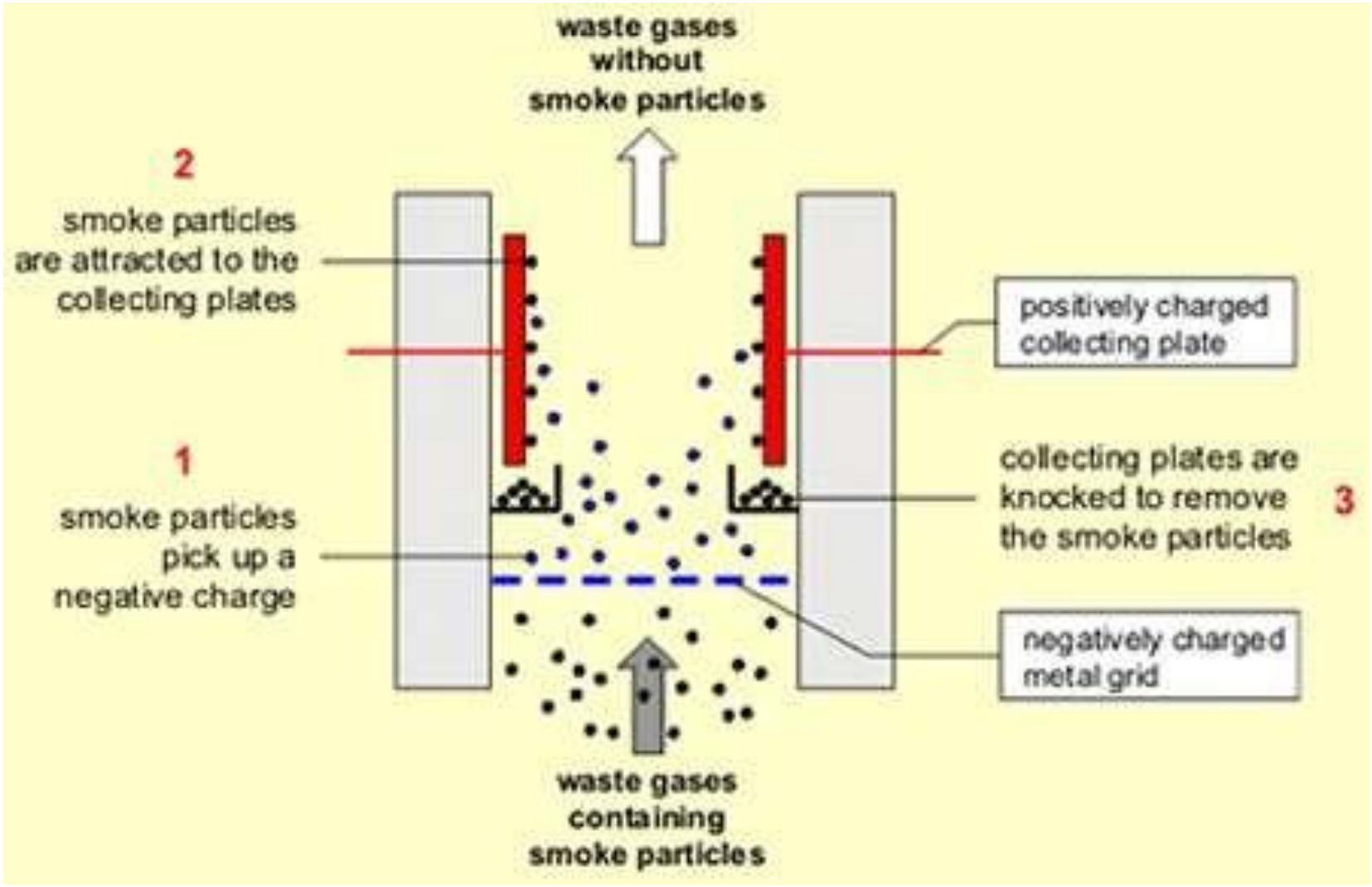


Figure: Electrostatic precipitator components

Conti….

- The particles are removed from the plates by "rapping" and collected in a hopper located below the unit.
- The removal efficiencies for ESPs are highly variable; however, for very small particles alone, the removal efficiency is about 99 percent.
- Electrostatic precipitators are not only used in utility applications but also other industries (for other exhaust gas particles) such as cement (dust), pulp & paper (salt cake & lime dust), petrochemicals (sulfuric acid mist), and steel (dust & fumes).


waste gases without smoke particles
2
smoke particles are attracted to the collecting plates
positively charged collecting plate
1
smoke particles pick up a negative charge
collecting plates are knocked to remove the smoke particles
3
negatively charged metal grid
waste gases containing smoke particles

Advantages

- High collection efficiency.
- Particles may be collected dry or wet.
- Can be operated at high temp. (300-450˚c).
- Maintenance is normal.
- Few moving parts.

Disadvantages

- High initial cost.
- Require high voltage.
- Collection efficiency reduce with time.
- Space requirement is more.
- Possible of explosion during collection of combustible gases or particulates.

Fabric filters

- Flue gas is allowed to pass through a woven Fabric, which filters out Particulate matter.
- Small particles are retained on the fabric.
- Remove particles up to 1 μm.
- Its efficiency up to 99%.



Fabric Filters

- Fabric filters, or baghouses, remove dust from a gas stream by passing the stream through a porous fabric. The fabric filter is efficient at removing fine particles and can exceed efficiencies of 99 percent in most applications.

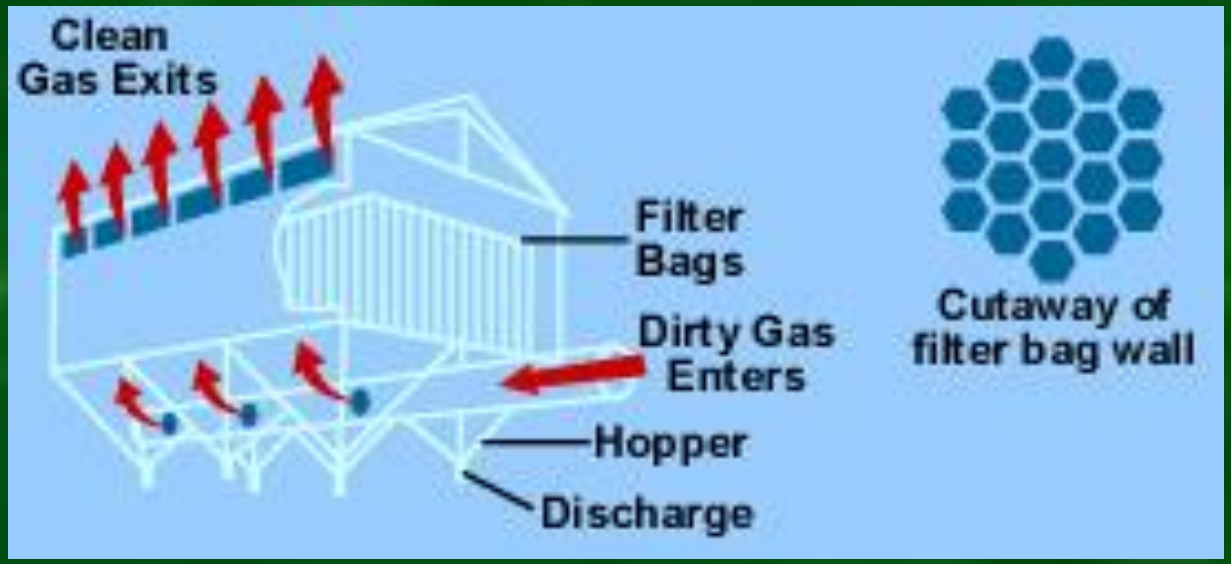


Figure: Fabric filter (baghouse) components

Advantages

- Higher collection efficiency for smaller than 10 μm particle size.
- Performance decrease becomes visible, giving prewarning.
- Normal power consumption.

Disadvantages

- High temp. gases need to be cooled.
- High maintenance and fabric replacement cost.
- Large size equipment.
- Fabric is liable to chemical attack.

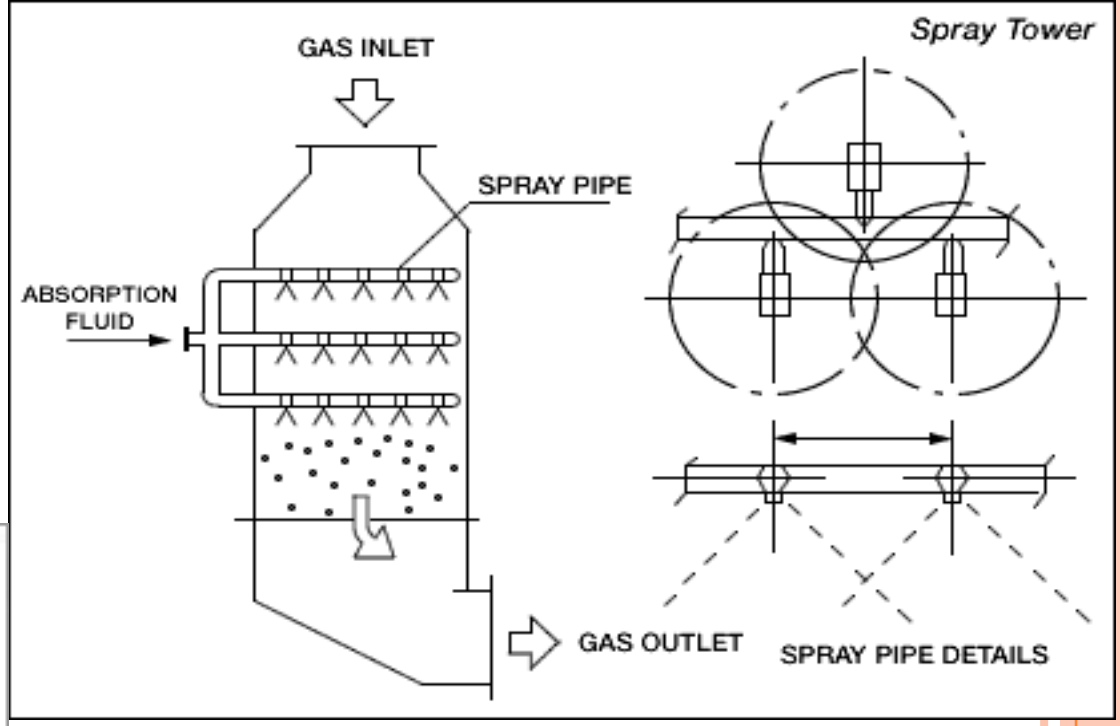
Wet Control Devices

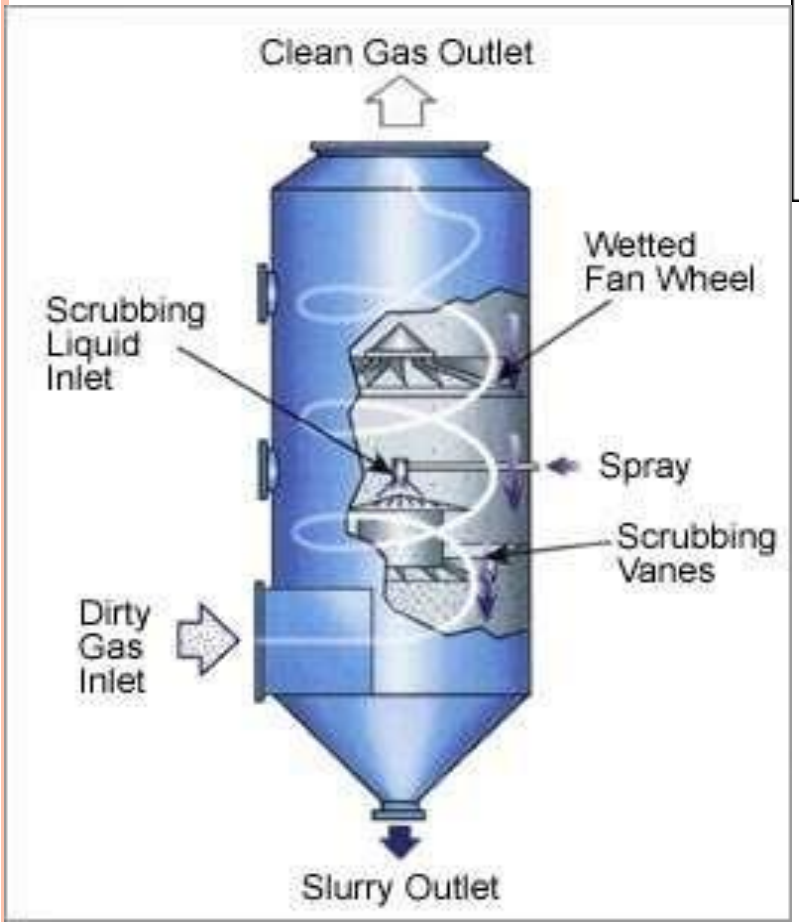
Scrubbers

- Particulate matters are incorporated into liquid droplets and removed from the gas stream.
- Different types of scrubbers are-
  - Spray tower
  - Venturi scrubber
  - Cyclone scrubber
- Flue gas made to push up against a down falling water current.
- Particulate matter mix up with water thus falls down and gets removed.



Spray tower





Cyclone scrubber

Advantages

- Simultaneously remove particulates and gaseous pollutants.
- Hot gases can be cooled down.
- Corrosive gases can be recovered and neutralize.

Disadvantages

- Lot of waste waters produced.
- Poses freezing problem in cold countries.
- Maintenance cost is high when corrosive materials are collected.

Venturi Scrubbers

- Venturi scrubbers use a liquid stream to remove solid particles.
- In the venturi scrubber, gas laden with particulate matter passes through a short tube with flared ends and a constricted middle.
- This constriction causes the gas stream to speed up when the pressure is increased.

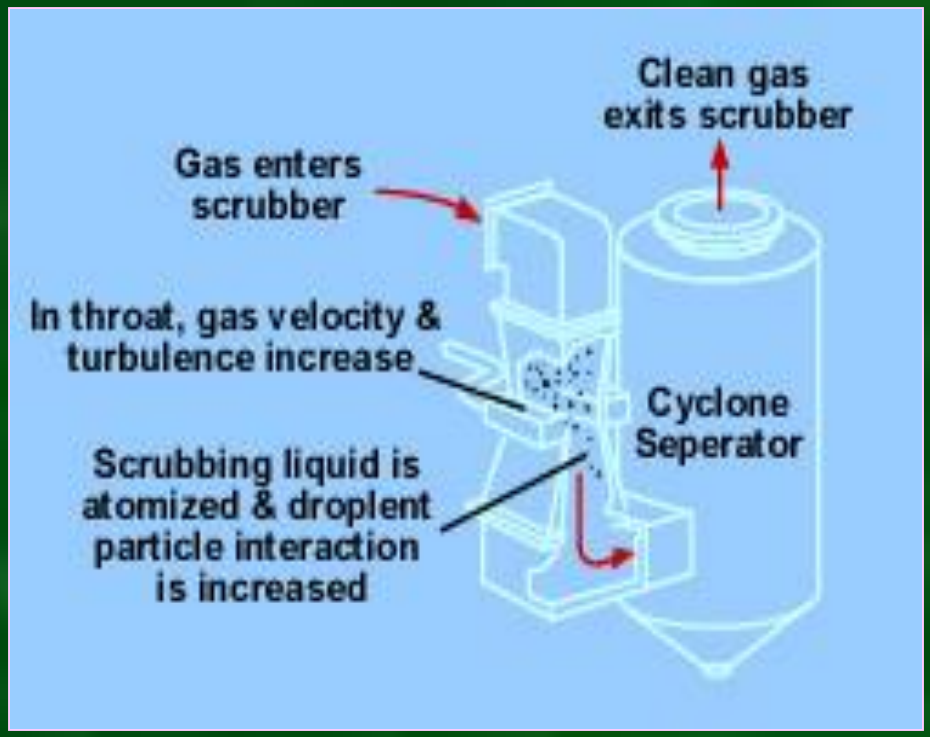


Figure: Venturi scrubber components

- The difference in velocity and pressure resulting from the constriction causes the particles and water to mix and combine.
- The reduced velocity at the expanded section of the throat allows the droplets of water containing the particles to drop out of the gas stream.
- Venturi scrubbers are effective in removing small particles, with removal efficiencies of up to 99 percent.
- One drawback of this device, however, is the production of wastewater.

Gaseous Control Technologies



Control of gaseous pollutants from stationary sources

- The most common method for controlling gaseous pollutants is the addition of add-on control devices to recover or destroy a pollutant.
- There are four commonly used control technologies for gaseous pollutants:
  - Absorption,
  - Adsorption,
  - Condensation, and
  - Incineration (combustion)

Absorption

- The removal of one or more selected components from a gas mixture by absorption is probably the most important operation in the control of gaseous pollutant emissions.
- Absorption is a process in which a gaseous pollutant is dissolved in a liquid.
- Water is the most commonly used absorbent liquid.
- As the gas stream passes through the liquid, the liquid absorbs the gas, in much the same way that sugar is absorbed in a glass of water when stirred.



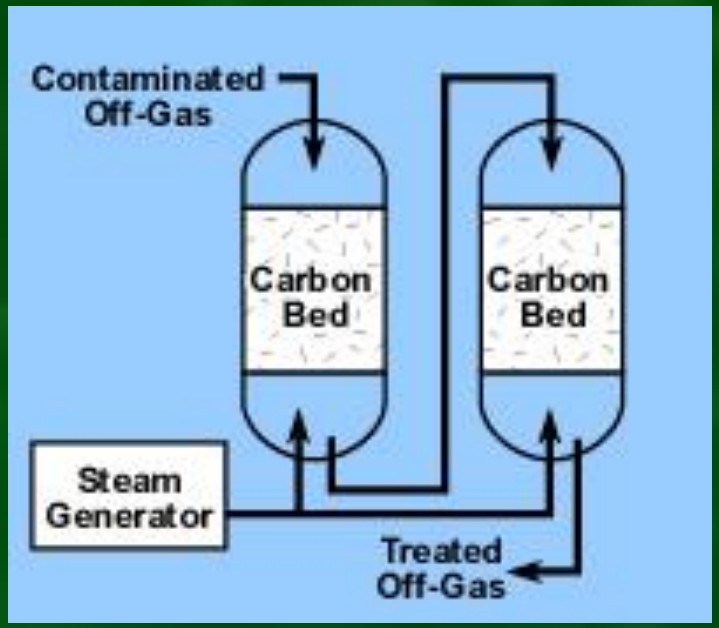
Typical Packed Column Diagram

- Absorbers are often referred to as scrubbers, and there are various types of absorption equipment.
- The principal types of gas absorption equipment include spray towers, packed columns, spray chambers, and venture scrubbers.
- In general, absorbers can achieve removal efficiencies grater than 95 percent. One potential problem with absorption is the generation of waste-water, which converts an air pollution problem to a water pollution problem.

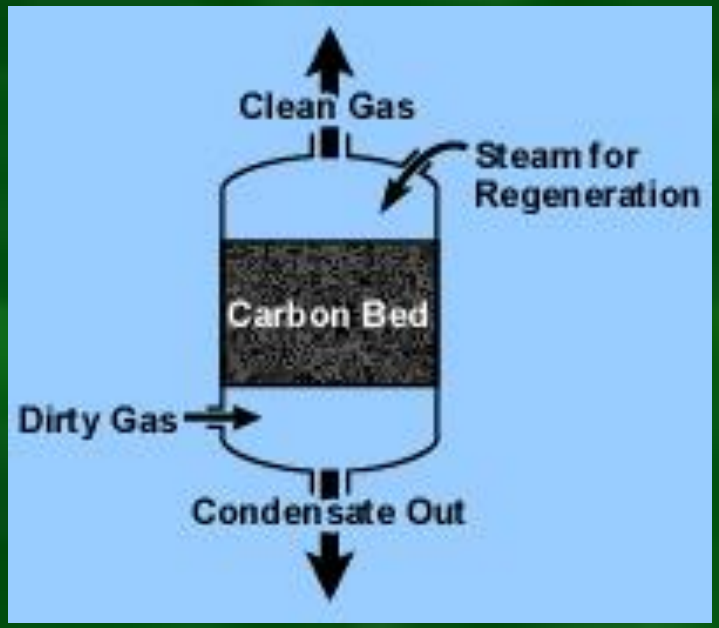
Adsorption

- When a gas or vapor is brought into contact with a solid, part of it is taken up by the solid. The molecules that disappear from the gas either enter the inside of the solid, or remain on the outside attached to the surface. The former phenomenon is termed absorption (or dissolution) and the latter adsorption.
- The most common industrial adsorbents are activated carbon, silica gel, and alumina, because they have enormous surface areas per unit weight.
- Activated carbon is the universal standard for purification and removal of trace organic contaminants from liquid and vapor streams.

- Carbon adsorption systems are either regenerative or non-regenerative.
  - **Regenerative system** usually contains more than one carbon bed. As one bed actively removes pollutants, another bed is being regenerated for future use.
  - **Non-regenerative systems** have thinner beds of activated carbon. In a non-regenerative adsorber, the spent carbon is disposed of when it becomes saturated with the pollutant.



Regenerative Carbon Adsorption System



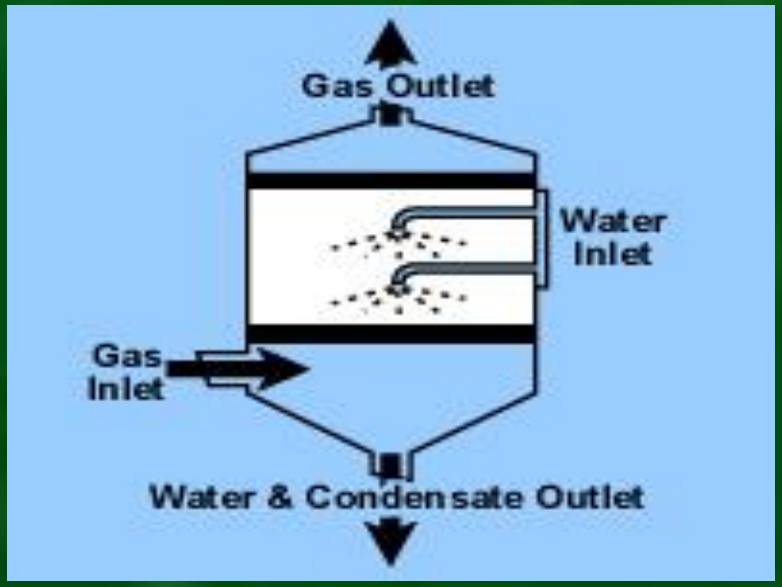
Non-Regenerative Carbon Adsorption System

Condensation

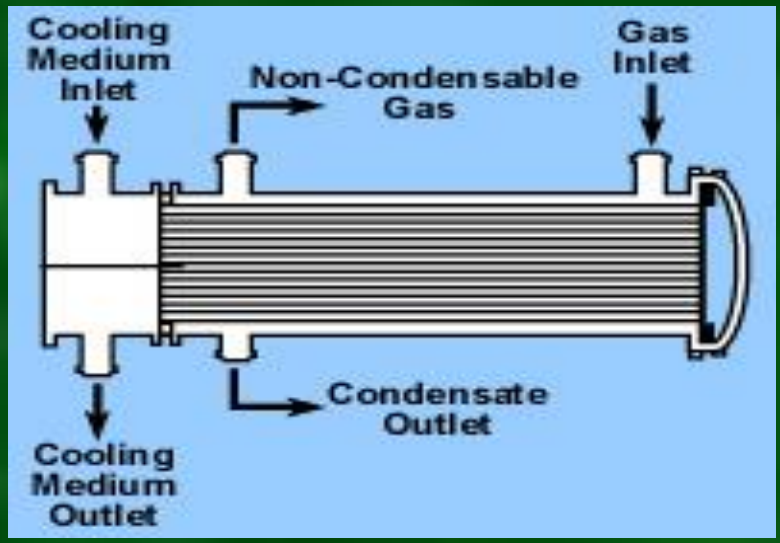
- Condensation is the process of converting a gas or vapor to liquid. Any gas can be reduced to a liquid by lowering its temperature and/or increasing its pressure.

- Condensers are typically used as pretreatment devices. They can be used ahead of absorbers, absorbers, and incinerators to reduce the total gas volume to be treated by more expensive control equipment. Condensers used for pollution control are contact condensers and surface condensers.

- **In a contact condenser**, the gas comes into contact with cold liquid.
- In a **surface condenser**, the gas contacts a cooled surface in which cooled liquid or gas is circulated, such as the outside of the tube.
- Removal efficiencies of condensers typically range from 50 percent to more than 95 percent, depending on design and applications.



Contact condenser



Surface condenser

Incineration

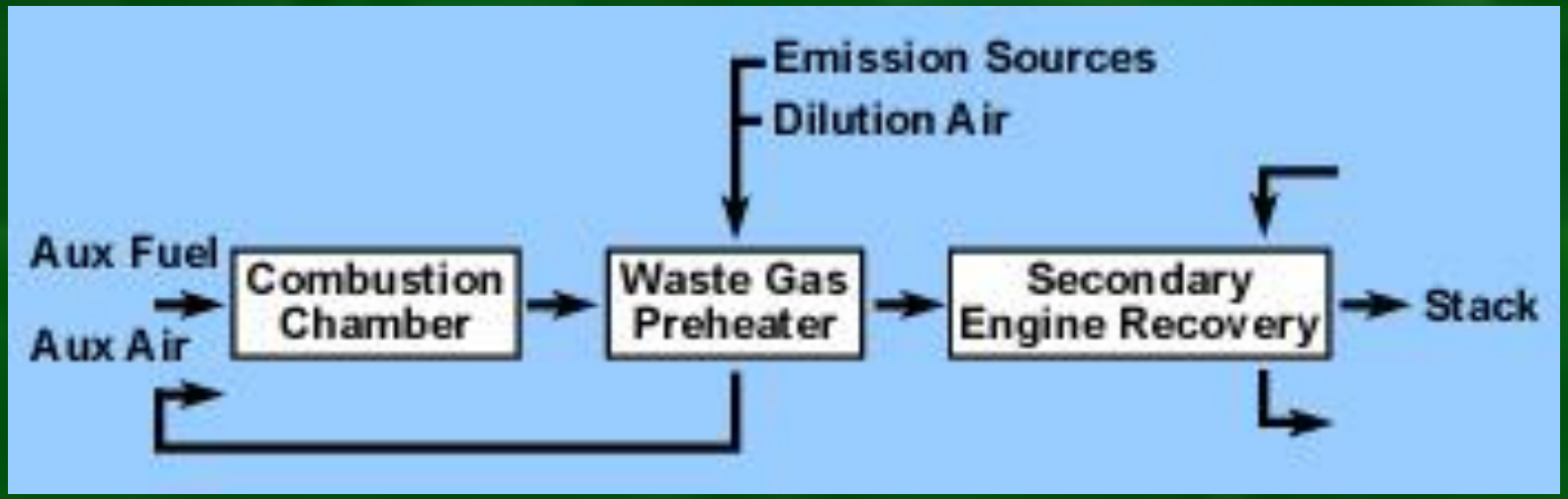
- Incineration, also known as combustion, is most used to control the emissions of organic compounds from process industries.
- This control technique refers to the rapid oxidation of a substance through the combination of oxygen with a combustible material in the presence of heat.
- When combustion is complete, the gaseous stream is converted to carbon dioxide and water vapor.
- Equipment used to control waste gases by combustion can be divided in three categories:
  - Direct combustion or flaring,
  - Thermal incineration and
  - Catalytic incineration.

Direct combustor

- **Direct combustor** is a device in which air and all the combustible waste gases react at the burner. Complete combustion must occur instantaneously since there is no residence chamber.
- A flare can be used to control almost any emission stream containing volatile organic compounds. Studies conducted by EPA have shown that the destruction efficiency of a flare is about 98 percent.

In **thermal incinerators** the combustible waste gases pass over or around a burner flame into a residence chamber where oxidation of the waste gases is completed.

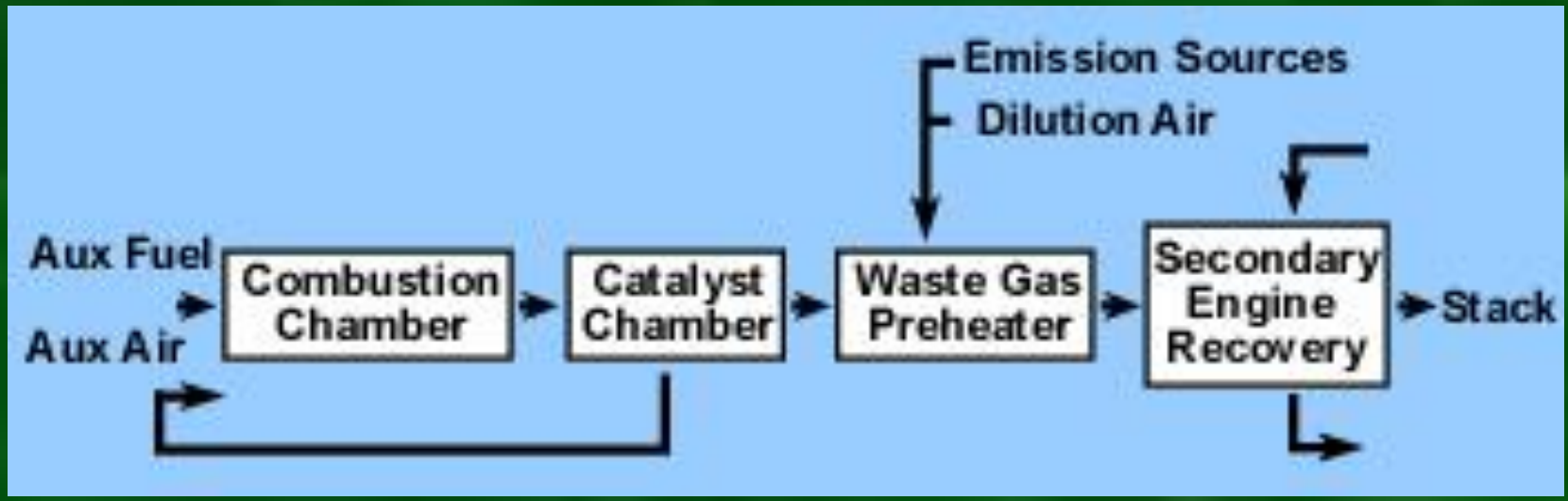
Thermal incinerators can destroy gaseous pollutants at efficiencies of greater than 99 percent when operated correctly.



Thermal incinerator general case

Catalytic incinerators are very similar to thermal incinerators. The main difference is that after passing through the flame area, the gases pass over a catalyst bed.

A catalyst promotes oxidation at lower temperatures, thereby reducing fuel costs. Destruction efficiencies greater than 95 percent are possible using a catalytic incinerator.



Catalytic incinerator

Air pollution dispersion modeling



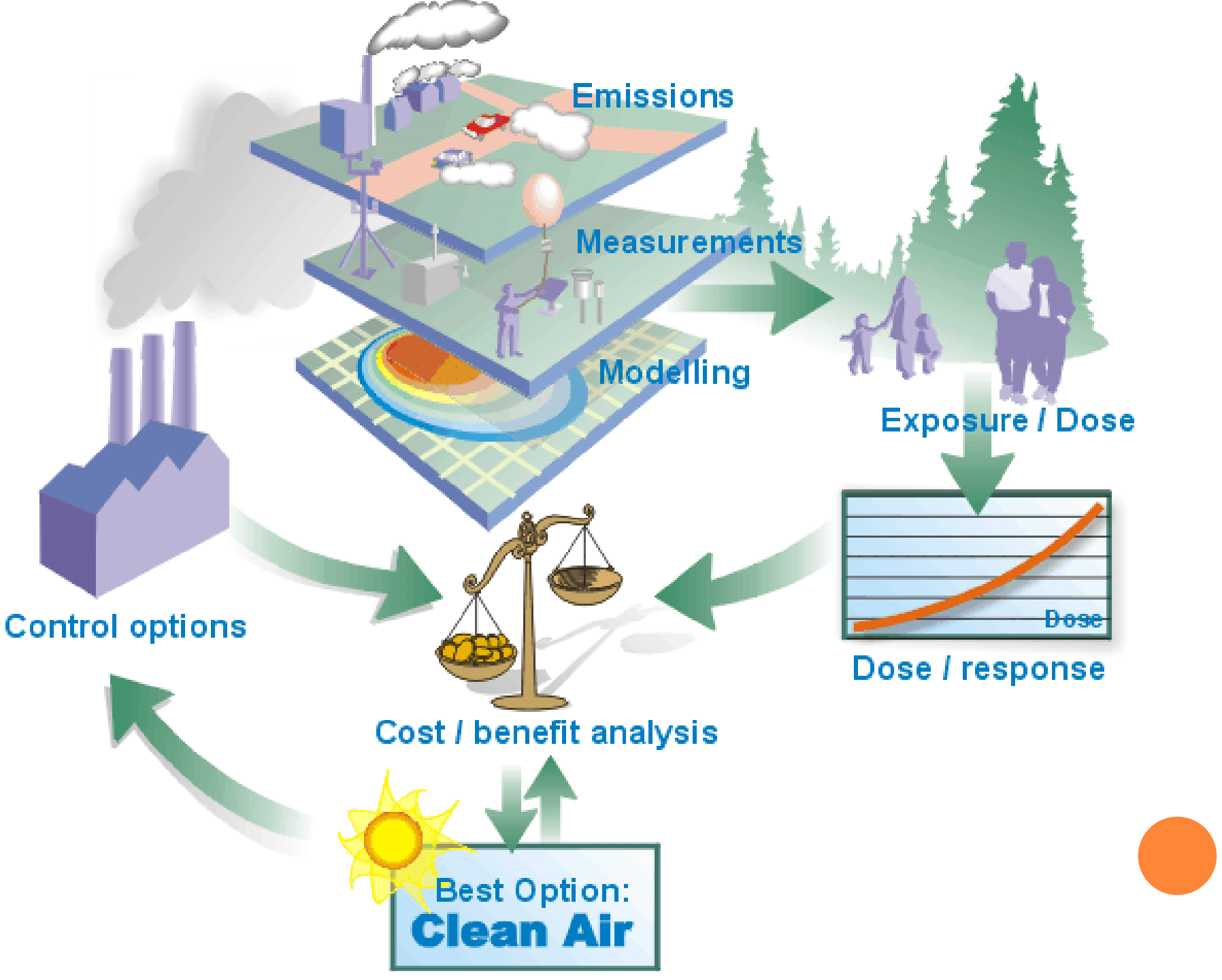
What is dispersion modeling

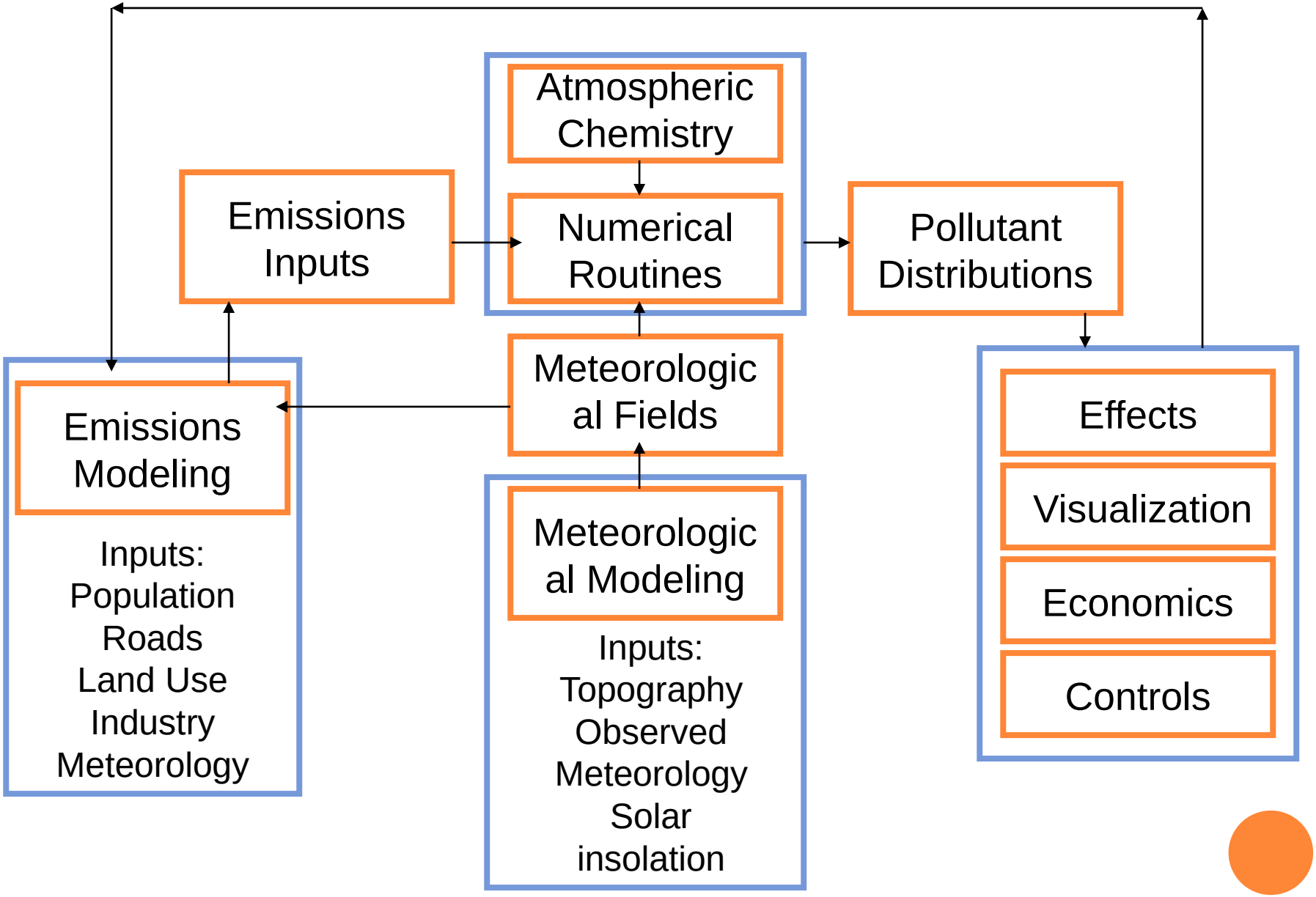
- It is an attempt to describe relationship between emission, occurring concentration and deposition
- It gives complete analysis of what emission sources have lead to concentration depositions
- Mathematical models use analytical and numerical formulations, usually implemented on computers



Why dispersion modeling is required

- To predict ambient air concentration which will result from a emission source
- To plan and execute air pollution control program considering cost effectiveness
- For environmental impact assessment
- Quantify the impact of process improvements
- Evaluating the performance of emission control techniques
- Optimization of stack height, diameter
- Planning the control of air pollution episodes


Emissions
Measurements
Modelling
Exposure / Dose
Dose
Dose / response
Control options
Cost / benefit analysis
Best Option:
Clean Air


Atmospheric Chemistry
Emissions Inputs
Numerical Routines
Pollutant Distributions
Meteorological Fields
Emissions Modeling
Inputs:
Population
Roads
Land Use
Industry
Meteorology
Meteorological Modeling
Inputs:
Topography
Observed Meteorology
Solar insolation
Effects
Visualization
Economics
Controls

Basics of air pollution dispersion models

- All air pollution models are based on the simple Material Balance Principles

- The general material balance equation for a air pollution model can be written as follows:

  Accumulation Rate = (All flow rates in)-(All flow rates out) + (Creation rate) – (Destruction Rate)



Input data required for dispersion models

1. Meteorological conditions
2. Emissions parameters
3. Terrain elevations at the source location and at the receptor location.
4. Details of obstructions if any



Types of air pollution dispersion models

- Box model
- Gaussian model
- Lagrangian model
- Eulerian model
- Dense Gas model

